A zinc–iodine flow-battery positive-electrode model coupling solid-iodine deposition and accessibility loss

Abstract

In zinc–iodine flow batteries (ZIFBs), the positive carbon-felt electrode couples soluble-iodine speciation, native-iodine deposition and progressive loss of electrochemically accessible area. We develop a calibrated multiscale positive-electrode scaffold that distinguishes free-I_2 saturation, retained solid-iodine inventory and functional accessibility. In the registered production branch at $40\,\mathrm{mA\,cm^{-2}}$ and in the representative $1\,\mathrm{M}$ ZnI_2+$4\,\mathrm{M}$ NH_4Br electrolyte, the model reaches its positive-electrode-averaged $S_{\mathrm{base}} = 1$ marker at $Q_{\mathrm{s}} = 83.0\,\mathrm{mAh\,cm^{-2}}$ and the voltage-calibrated half-accessibility marker at $Q_{\mathrm{f,cal}} = 99.6\,\mathrm{mAh\,cm^{-2}}$; smooth relative permeability remains 0.956 at $120\,\mathrm{mAh\,cm^{-2}}$. Lower-scale calculations constrain placement, transport and physical closure families rather than fixing calibrated accessibility. In fresh matched true-mesh solves, replacing only the production accessibility relation with the dense physical-island relation changes Q_{s} by only $0.002\,\mathrm{mAh\,cm^{-2}}$, but changes the endpoint accessibility loss from 0.990 to 0.309 and the voltage from $1.728\,\mathrm{V}$ to $1.440\,\mathrm{V}$ ($-288.2\,\mathrm{mV}$); the physical-island case does not reach half accessibility by $120\,\mathrm{mAh\,cm^{-2}}$. This divergence shows that accessibility feedback changes the solid-inventory and voltage trajectories, while voltage alone does not identify the deposit morphology. Existing single-cell composition, rate and thickness data provide protocol-bounded context, with NH_4Br serving only as a representative supporting electrolyte. The framework separates free-iodine saturation, native-solid deposition and closure-dependent accessibility loss, and identifies the transport and morphology parameters that move the modeled positive-electrode operating window.

Keywords: zinc–iodine flow battery; positive electrode; iodine super-saturation; accessibility loss; porous-electrode model; operating boundary; multiscale modeling

1 Introduction

Zinc–iodine flow batteries (ZIFBs) are an attractive chemistry for low-cost, safe aqueous grid storage: they pair an abundant, high-potential iodide/iodine catholyte with a zinc anode, decouple power from energy, and reach high theoretical capacity, with recent cells pushing toward practical high areal loadings [1–7]. Realizing that promise at high loading and high current, however, is limited less by the redox couples themselves than by what the charged phases do inside the porous electrodes.

A practical ZIFB has several coupled failure channels. On the negative side, non-uniform zinc plating forms dendrites and a compact zinc layer near the electrode–membrane interface that starves the porous felt of ions and can perforate the separator [8–10]. In the electrolyte, dissolved polyiodide can shuttle across the membrane and self-discharge the cell [11–13]. On the positive side, the iodine redox must run through a poorly soluble neutral intermediate: charging oxidizes iodide to I_2 ($2\,I^- \rightarrow I_2 + 2e^-$), which iodide immediately binds into soluble triiodide ($I^- + I_2 \rightleftharpoons I_3^-$, $K \approx 698\,\mathrm{M^{-1}}$ [4, 14]). Most of the oxidized iodine is thus *hidden* in a soluble complex, so precipitation is governed not by the total oxidized inventory but by the small *free* I_2 monomer left over: once that free monomer outruns the buffer it can precipitate as native $I_2(s)$. Independent electrochemical and confined-host studies resolve or retain solid iodine under their own electrode conditions [15, 16]; they do not establish the deposit geometry in the present carbon felt.

The negative-side and shuttle channels have received sustained mechanistic attention, but the positive iodine electrode is less resolved. Its failure is subtler than a solubility limit. Retained $I_2(s)$ is not, by itself, the failure: a solid can sit in the felt without occluding the percolating reaction and transport network; it becomes limiting only once its morphology overlaps the active area and the ion/mass-transfer paths. Both sides of that distinction have host-specific precedents. In-situ optical microscopy in a distinct dual-plating, static graphite-felt iodine chemistry resolves a continuous I_2 film during charge and discrete residual particles after discharge [17]. A nanoconfined activated-carbon platform reports at least about 30% iodine pore filling while retaining high coulombic efficiency and cycling stability [16]; this is a host-bounded counterexample, not a felt threshold. A dense iodine film can raise positive-electrode charge-transfer resistance roughly ninefold [18]. Dissolution measurements also show that solvent environment changes the solid-iodine clearance scale and high-rate response [19]. The unresolved modeling question is therefore not only the *appearance* of iodine but the state relation by which retained inventory becomes associated with loss of electrochemically accessible area. Continuum porous-electrode theory is well established for this kind of question—from the classical framework of Newman and co-workers [20] to modern tertiary-current-distribution solvers for flow cells [21–23], and to reduced-order models of solid-product passivation in Li–S and related electrodes [24]—but none of these resolves, for the iodine positive electrode, the specific map from retained native $I_2(s)$ inventory to *functional* accessibility and transport loss. For that transition, an auditable, conditional electrode-resolved scaffold that separates an averaged saturation marker from morphology-dependent accessibility loss remains missing.

The supporting electrolyte here is NH_4Br. It is treated as a representative formulation context rather than the paper's protagonist: the modeled positive-electrode branch is I_2/I^-, not bromide oxidation. Composition-dependent speciation and transport require separate measurements and are therefore discussed descriptively (§2.8, §5.3).

Here we build a calibrated iodine-state scaffold for the ZIFB positive electrode (Fig. 1). The continuum model supplies spatial fields of free-iodine stress and retained solid fraction. DFT supplies a relative placement tendency at oxygenated carbon sites, not an absolute nucleation density; molecular dynamics supplies a bounded bulk-electrolyte carrier-diffusivity prior; the single-fibre calculations define isolated-island accessibility families; and the pore-network calculations bound the smooth bulk-permeability response. The effective accessibility magnitude used by the production continuum branch is calibrated to the voltage response and is therefore kept distinct from the physical isolated-island closure (Fig. 2). We organize the resulting charge trajectory as a state clock in areal capacity Q and project it into a positive-electrode J–Q map. Published iodine-dissolution measurements provide condition-specific external kinetic scales through $i^* = 2Fk$ [19]; they do not measure a universal current ceiling for the present electrolyte. The model resolves one galvanostatic charge and does not establish cycle-to-cycle reset, discharge recovery or a unique deposit morphology. Existing compression, composition and rate data are reported with their source-specific roles as descriptive external-consistency evidence; they do not independently validate the production thresholds.

Solid reaction products passivate or disconnect the active surface in other aqueous and condensed electrodes, from ZnO to Li_2S and MnO_2 [25–27]; the iodine case differs mainly in lacking such a porous-electrode map, and it is that map this work supplies.

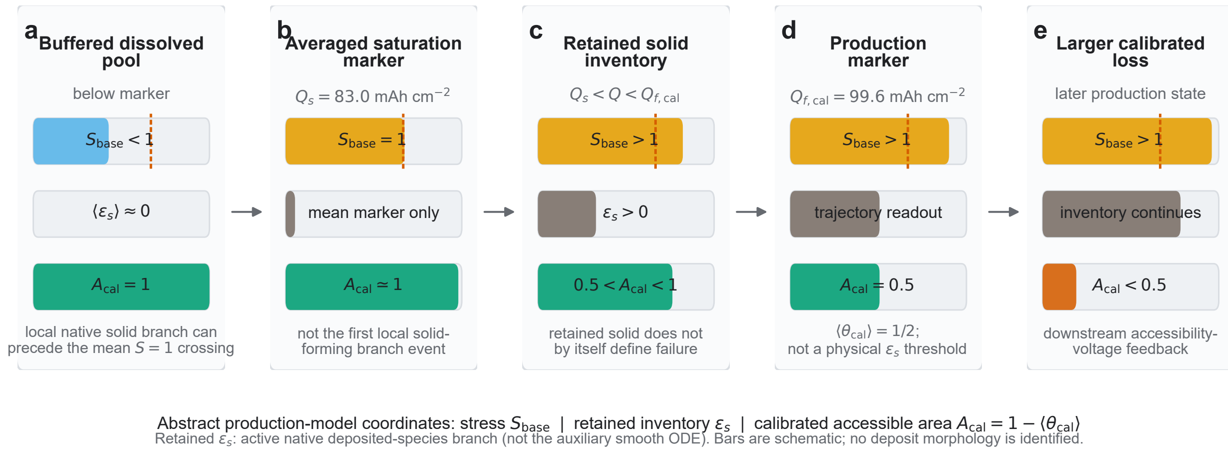

Figure 1: Conceptual iodine-state progression in the ZIFB positive electrode. (a) Before the registered averaged marker, the mean oxidized-iodine inventory remains dominated by the dissolved, buffered pool; local native-solid-path activation can precede that mean crossing. (b) The production model defines its registered averaged-saturation marker by $S_{\mathrm{base}} = 1$. (c) Beyond the registered saturation marker, retained $I_2(s)$ can accumulate while substantial reaction area remains accessible. (d) The production branch defines its functional marker by $\langle \theta_{\mathrm{cal}} \rangle = \frac{1}{2}$, not by the physical-island or film-percolation solid fraction. (e) Larger effective accessibility loss is a downstream modeled state. At the registered baseline, $Q_s = 83.0\,\mathrm{mAh\,cm^{-2}}$ and $Q_{\mathrm{f,cal}} = 99.6\,\mathrm{mAh\,cm^{-2}}$. The artwork defines model states; it is not an image of the deposit and does not identify a continuous film or local pore-throat blockage. The scale-to-variable evidence types are summarized in Fig. 2.

2 Model framework and reduced iodine-state theory

The positive-electrode scaffold follows one state chain: free-I_2 super-saturation S_{base} drives retained solid fraction ε_{s}, and an accessibility closure maps ε_{s} to loss of electrochemically reachable area. The registered production branch defines the averaged-saturation marker Q_{s} by $S_{\mathrm{base}} = 1$ and a calibrated half-accessibility marker $Q_{\mathrm{f,cal}}$ by $\langle \theta_{\mathrm{cal}} \rangle = \frac{1}{2}$. Physical isolated-island and film-percolation thresholds are separate comparators. This section defines the state chain and its closure-conditioned operating boundary; the continuum implementation is deferred to §3. Charging oxidizes iodide, $2\,I^- \to I_2 + 2e^-$, but a complexation buffer $\beta = 1 + K_{\mathrm{I}} c_{\mathrm{I}^-} + K_{\mathrm{Br}} c_{\mathrm{Br}^-}$ holds the *free* I_2 concentration $c^{\mathrm{f}}_{I_2} = c_{I_2,\mathrm{tot}}/\beta$ below the effective saturation $c^{\mathrm{eff}}_{\mathrm{sat}}$. The super-saturation $S = c^{\mathrm{f}}_{I_2}/c^{\mathrm{eff}}_{\mathrm{sat}}$, not the total oxidized inventory, therefore defines the model's operational saturation coordinate. Local regions can cross $S = 1$ and activate the native-solid branch before the electrode average reaches unity, so Q_{s} is not the first local solid. Three state variables then carry the model: S_{base}, ε_{s} and a named accessibility loss—θ_{cal} for production or θ_{island} for the physical-island branch. The production interval $\Delta Q_{\mathrm{cal}} = Q_{\mathrm{f,cal}} - Q_{\mathrm{s}}$ is therefore a closure-conditioned state-clock output (Fig. 1b–d). The accessible fraction is $A_{\mathrm{bare}} = 1 - \theta$; we avoid using "coverage" as a morphology label because voltage does not identify the deposit geometry. The free-I_2 super-saturation that produces the solid is itself a *mass-transfer* stress, set by the oxidized-carrier diffusivity, the boundary layer, and the accommodation chemistry. A slower-diffusing electrolyte therefore moves the boundary through transport rather than through accessibility alone; viscosity acts separately through the Darcy-pressure channel unless it also changes diffusivity. Because the generation load rises with current density J and the inventory with capacity Q, S is a function of the operating point (J, Q). The saturation and calibrated-accessibility markers become two curves in the J–Q plane. Published dissolution mea-

Table 1: Principal symbols and baseline values (bulk electrolyte 1 M ZnI_2 + 4 M NH_4Br, i.e. bulk $c_{I^-} = 2$ M and $c_{Br^-} = 4$ M; $i_{app} = 400\,A\,m^{-2}$, $T = 293.15$ K in the registered continuum solve). The buffer β (and hence S_{base}) is evaluated at the *current-on surface-effective* iodide, not the bulk value; this boundary-layer reduction is derived in §2.1. Reused letters are kept distinct: the dissolution constant k ($i^* = 2Fk$), the accessibility prefactor k_{geo}, and the complexation constants K_I, K_{Br} are different quantities.

symbol	meaning	value / units
n, F	electrons per I_2; Faraday	2; 96 485 $C\,mol^{-1}$
$c_{I_2,tot}, c^f_{I_2}$	total / free dissolved I_2	$mol\,m^{-3}$
c_{sat}	I_2 saturation reference	1.33 $mol\,m^{-3}$
β	complexation buffer $1 + K_I c_{I^-} + K_{Br} c_{Br}$	554–1263 (surface)
D_0, D_{eff}	free-solution / Bruggeman oxidized-carrier diffusivity (lumped $I_3{}^-/I_2Br^-$; the neutral-I_2 prior is 0.6×10^{-9}, SI)	0.5×10^{-9}; $D_0\varepsilon^{1.5}$ $m^2\,s^{-1}$
δ	Nernst-layer thickness	25×10^{-6} m
$\varepsilon, \varepsilon_s$	pore-liquid / solid-I_2 fraction	$\varepsilon = \varepsilon_0 - \varepsilon_s$
V_m	molar volume of I_2(s)	$5.15 \times 10^{-5}\,m^3\,mol^{-1}$
Q	areal capacity (charge clock)	$mA\,h\,cm^{-2}$ ($= 3.6 \times 10^4\,C\,m^{-2}$)
S	super-saturation stress $c^f_{I_2}/c^{eff}_{sat}$	dimensionless
S_{base}	within-cycle baseline stress	$\gamma_{salt}(\Pi_{inv} + \Pi_{gen})/\beta_{intra}$
θ_{cal}	production voltage-calibrated accessibility loss	$Q_{f,cal} : \langle\theta_{cal}\rangle = \frac{1}{2}$
θ_{island}	physical isolated-island accessibility loss	$1 - \exp(-35.4\varepsilon_{s,reg}^{0.6222})$
A_{bare}	accessible carbon fraction in the active closure	$1 - \theta$
N_{I_2}	dynamic dimensionless state in the production accessibility relation	solved state
N	swept areal-density prior in the physical single-fibre family	m^{-2}
35.4, 0.6222	fitted dense-island coefficient and exponent (Eq. 2)	dimensionless
$\varepsilon^*_{s,cal,traj}$	mean production solid fraction at $Q_{f,cal}$	1.18572×10^{-4}
$\varepsilon_{s,island,1/2}$	dense-island half-accessibility fraction	1.7973×10^{-3}
$\varepsilon^*_{s,film\text{-}perc}$	geometric connected-film/percolation marker	2.2324×10^{-3}
ϕ_{ppt}	reduced solid-formation efficiency per unit overshoot $(S-1)_+$	dimensionless; closure-cond.
$i^* = 2Fk$	condition-labelled dissolution-derived current scale	$A\,m^{-2}$

surements provide the condition-specific scale $i^* = 2Fk$, rather than a universal ceiling for this electrolyte. The rest of this section makes that picture quantitative, building in turn the reduced one-carrier scaffold (§2.1), the closures that fix its coefficients (§2.2), the super-saturation stress (§2.3), and the resulting J–Q boundary with its production markers and external kinetic scale (§2.5–2.7), before separating supporting-electrolyte context (§2.8). The explicit continuum governing balances and their assumptions are deferred to the implementation (§3.1); symbols are collected in Table 1.

2.1 State variables and the reduced scaffold

We write the reduced criterion along the through-plane coordinate of the volume-averaged positive felt, $x \in [0, L]$ (thickness $L \approx 2$ mm; collector at $x = 0$, separator at $x = L$); the full continuum solve of §3 is two-dimensional, retaining the flow-wise coordinate y and labeling x in the cell frame (3–5 mm) in the field maps (Fig. 4 and Supplementary Fig. S8). The felt carries the overall couple

$2\,\mathrm{I}^- \rightleftharpoons \mathrm{I}_2 + 2\,\mathrm{e}^-$ through two numerical reaction paths: a bare-carbon soluble-iodine path and a super-saturation-activated native-solid path. The continuum scaffold (§3; SI) transports one *lumped* oxidized-iodine pool O ($\mathrm{I_3}^-$, $\mathrm{I_2Br}^-$ and aqueous $\mathrm{I_2}$ in fast complexation equilibrium). The remaining free-iodide inventory is reconstructed stoichiometrically from this solved field, rather than advanced by a second species equation. Because bromide is in large excess ($4\,\mathrm{M}\,\mathrm{NH_4Br}$), explicit species electromigration and a multi-ion electroneutrality closure are not resolved. Supporting-electrolyte charge transport is represented by the prescribed effective liquid conductivity κ_l, whereas bromide enters the free-iodine stress through the complexation factor β. The resulting iodide coordinate depletes with the solved oxidized pool but is an algebraic stoichiometric inventory, not an independently transported field. This is a reduced one-carrier scaffold, not a multi-ion Nernst–Planck solve. Two ingredients carry the mechanism. First, the neutral intermediate is *buffered*: fast complexation partitions the total dissolved iodine $c_{\mathrm{I_2,tot}} = c^{\mathrm{f}}_{\mathrm{I_2}} + c_{\mathrm{I_3}^-} + c_{\mathrm{I_2Br}^-}$ instantaneously into a precipitable free monomer,

$$c^{\mathrm{f}}_{\mathrm{I_2}} = \frac{c_{\mathrm{I_2,tot}}}{\beta}, \qquad \beta = 1 + K_{\mathrm{I}}\, c_{\mathrm{I}^-} + K_{\mathrm{Br}}\, c_{\mathrm{Br}}, \tag{1}$$

so β is the factor by which complexation inflates the soluble inventory above the precipitable monomer. At the current-on surface it falls from $\beta_{\mathrm{surf}} \approx 1263$ at charge initiation toward $\beta \approx 554$ as iodide is consumed on charge. The initial value follows from the boundary layer: the current-on oxidized-iodine build-up ($c_{\mathrm{I_2,surf}} \approx 104\,\mathrm{mol\,m^{-3}}$ at $t = 0^+$) draws the free iodide at the reaction front from the bulk 2000 down to $\approx 1688\,\mathrm{mol\,m^{-3}}$ (each $\mathrm{I_2}$ sequesters ≈ 3 iodide via $\mathrm{I_3}^-$), giving $\beta_{\mathrm{surf}} \approx 1263$ against a fresh-bulk $\beta_0 \approx 1487$. (The reduction reads β at the running surface iodide; freezing it at the initial value is a stated approximation that shifts S but not its ordering.) The production solves adopt $K_{\mathrm{I}} = 720$ and $K_{\mathrm{Br}} = 11.5\ \mathrm{M}^{-1}$ (SI, central parameters), within the literature spread of the constants quoted in the text (698 and 12.2 M^{-1}); the β endpoints above are the solver values. Second, accessibility gates the bare-carbon Butler–Volmer area, while a separate native-solid Tafel path is activated by local free-iodine super-saturation. Their current partition advances the native deposited-species inventory and produces the accessibility–voltage non-identifiability analyzed below. The pore-phase balance, two-phase charge conservation, the Bruggeman corrections $D_{\mathrm{eff}} = D_0\varepsilon^{1.5}$, and the auxiliary nucleation-state ODE are stated explicitly in §3.1 (full implementation in the SI). Together with the closures of §3, they form the continuum iodine-state scaffold, which we now reduce to a single surface stress that the scaffold's averaged super-saturation reproduces up to a linear normalization. The scaffold deliberately does *not* resolve discrete pore-throat clogging, explicit multi-ion migration, the pump/pressure hydrodynamics, the collector-side shell, or the noise-level failure-state impedance. These lie outside the present reduction and enter only as bounds or external checks.

2.2 Multiscale priors and closures

The lower scales constrain different parts of the positive-electrode scaffold at different evidence levels. DFT constrains relative iodine placement at carbon sites; it does not determine a nucleation density. Molecular dynamics bounds bulk-electrolyte carrier diffusivities; it does not simulate transport through a deposited iodine film. The single-fibre calculations define physical isolated-island accessibility families, whereas the pore-network calculation bounds the smooth permeability response. For the fitted dense-island family,

$$\theta_{\mathrm{island}}(\varepsilon_{\mathrm{s,reg}}) = 1 - \exp\!\left(-35.4\,\varepsilon_{\mathrm{s,reg}}^{0.6222}\right), \qquad \varepsilon_{\mathrm{s,island},1/2} = 1.7973 \times 10^{-3}. \tag{2}$$

The production continuum branch uses a distinct effective relation, $\theta_{\mathrm{cal}} = 1 - \exp[-6N_{\mathrm{I_2}}^{1/3}\varepsilon_{\mathrm{s,reg}}^{2/3}]$, in which $N_{\mathrm{I_2}}$ is a dynamic dimensionless state. Its magnitude is voltage-calibrated and its declared

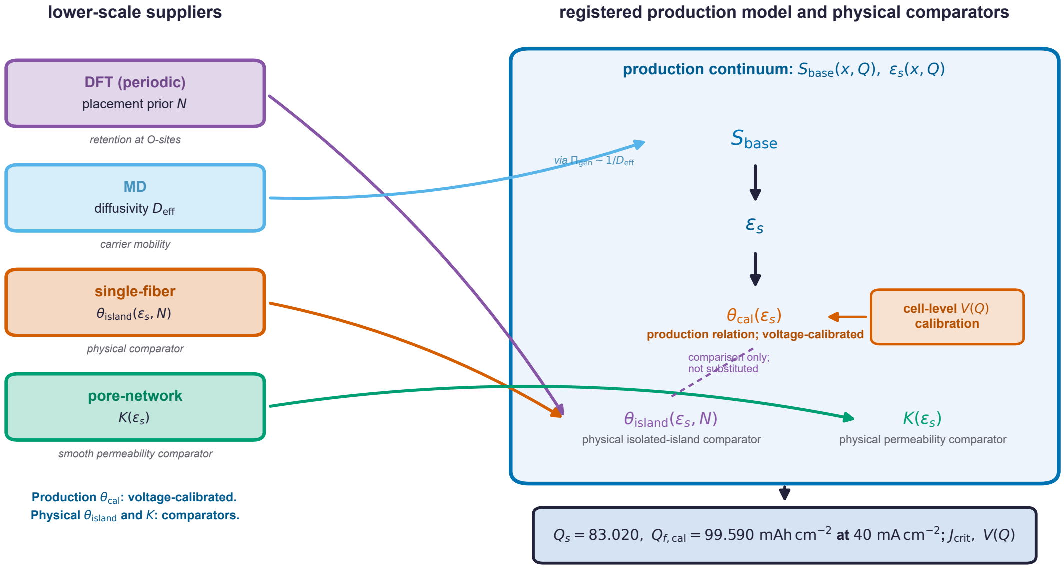

Figure 2: Evidence provenance across scales. DFT supplies a relative placement bias; molecular dynamics supplies a bounded bulk-carrier diffusivity prior; the single-fibre model supplies physical isolated-island accessibility families; the pore-network model supplies a smooth relative-permeability closure; and the continuum model supplies the spatial iodine-state fields. The production accessibility magnitude is voltage-calibrated. Arrows are labelled by evidence type—solved field, bounded prior, physical closure or calibrated effective closure—and do not imply a parameter-free bottom-up prediction.

midpoint occurs on the production trajectory at $\varepsilon^*_{\mathrm{s,cal,traj}} = 1.18572 \times 10^{-4}$. The physical dense-island midpoint and the geometric film-percolation marker $\varepsilon^*_{\mathrm{s,film\text{-}perc}} = 2.2324 \times 10^{-3}$ are separate comparators. We therefore propagate θ_{cal}, θ_{island} and $K(\varepsilon_{\mathrm{s}})$ as distinct objects. The composition factors introduced later are descriptive cell-level coordinates rather than microscopic constants.

2.3 Free-iodine balance and the super-saturation stress

At the carbon reaction front, the quasi-steady reduced balance across a Nernst layer of thickness δ balances the charge-equivalent supply $g = i_{\mathrm{app}}/(2F)$ against diffusive removal, $D_{\mathrm{eff}}\,\Delta c^{\mathrm{f}}_{\mathrm{I_2}}/\delta = g/\beta$ across the layer. Combining this balance with Eq. (1) gives the surface free-iodine concentration

$$c^{\mathrm{f}}_{\mathrm{I_2}} \;=\; \frac{1}{\beta}\left[\, c_{\mathrm{I_2,tot}} \;+\; \frac{i_{\mathrm{app}}\,\delta}{2F\,D_{\mathrm{eff}}} \right], \tag{3}$$

and the local super-saturation is $S = c^{\mathrm{f}}_{\mathrm{I_2}}/c^{\mathrm{eff}}_{\mathrm{sat}}$ (referenced to the effective, salting-out solubility $c^{\mathrm{eff}}_{\mathrm{sat}} = c_{\mathrm{sat}}/\gamma_{\mathrm{salt}}$; §2.6). The reduction neglects the late bare-versus-native-solid current partition; the native-solid path carries only 0.552% of integrated charge, although its instantaneous share rises near the endpoint (SI). Here $c_{\mathrm{I_2,tot}}$ denotes the *local* dissolved oxidized-iodine pool at the reaction front: the surface value of the solved continuum field, not the cell-averaged inventory. When the reduced expression is projected onto a capacity axis, it uses the solved surface trajectory, so the two terms of the bracket sit at the same location. (The balance $D_{\mathrm{eff}}\Delta c^{\mathrm{f}}_{\mathrm{I_2}}/\delta = g/\beta$ also takes β as uniform across the thin layer, consistent with the fast complexation assumption.) The bracket in Eq. (3) is exactly the surface variable carried by the continuum model (§3): the first term is the accumulated oxidized-iodine inventory, the second is the generation-driven boundary-layer build-up. This buffered form also encodes why solid I_2 forms late in the production trajectory: in an iodide-rich electrolyte the oxidation product is buffered as soluble $I_3{}^-$, so the modeled free-I_2 stress increases as the local iodide coordinate and hence β decrease. The buffer $\beta = \beta(c_{I^-})$ in Eq. (3) carries this model dependence through the solved surface-effective iodide. This is an internal state relation; the cited electrodeposition literature documents solid iodine formation but does not verify local iodide depletion in the present ZIFB [28].

2.4 Dimensionless groups and the baseline stress

Dividing Eq. (3) by c_{sat} defines three dimensionless groups,

$$\Pi_{\mathrm{gen}} = \frac{i_{\mathrm{app}}\,\delta}{2F\,D_{\mathrm{eff}}\,c_{\mathrm{sat}}}, \qquad \Pi_{\mathrm{inv}} = \frac{c_{I_2,\mathrm{tot}}}{c_{\mathrm{sat}}}, \qquad \beta\ (\text{Eq. 1}), \tag{4}$$

so that the *within-cycle* baseline super-saturation stress—the load-bearing coordinate of the whole framework—is

$$\boxed{S_{\mathrm{base}} = \gamma_{\mathrm{salt}}\,\frac{\Pi_{\mathrm{inv}} + \Pi_{\mathrm{gen}}}{\beta_{\mathrm{intra}}(Q)} = \frac{c^{\mathrm{f}}_{I_2}}{c^{\mathrm{eff}}_{\mathrm{sat}}}, \qquad \beta_{\mathrm{intra}} \equiv \beta\ (\text{Eq. 1}),} \tag{5}$$

read at the running iodide; this is the coordinate the continuum model solves and that the two-threshold, compression, and rate analyses below all use (throughout, "S" denotes S_{base} at the modeled baseline electrolyte). Here Π_{gen} is a generation/transport Damköhler number (in the spirit of Damköhler-type dimensionless reductions of flow-cell behavior [29]): the current density (rate) and the effective diffusivity—and hence *compression*, through the selected Bruggeman-type empirical effective-medium closure $D_{\mathrm{eff}} = D_0\varepsilon^{1.5}$—enter only here. The historical Bruggeman framework [30] does not measure this exponent for the compressed graphite felt. We quote the baseline reference value at the free-solution reference ($D_{\mathrm{eff}} \to D_0$, the Nernst layer sitting in the free electrolyte), $\Pi_{\mathrm{gen}} \approx 78$, so that compression enters as the $\varepsilon^{1.5}$ modifier rather than being folded into the baseline number. The inventory group Π_{inv} carries the state of charge and the iodide inventory, and the buffer β falls as iodide is consumed during charge (its *within-cycle* role, which the continuum solve carries). At the modeled baseline, the registered averaged-saturation marker is $\langle S_{\mathrm{base}}\rangle_{\mathrm{aveop1}} = 1$, where the corresponding averaged free-I_2 stress reaches the effective (salting-out) reference $c^{\mathrm{eff}}_{\mathrm{sat}} = c_{\mathrm{sat}}/\gamma_{\mathrm{salt}}$ (§2.6); the factor γ_{salt} in Eq. (5) converts the intrinsic-solubility groups Π to this effective reference, so $S_{\mathrm{base}} = c^{\mathrm{f}}_{I_2}/c^{\mathrm{eff}}_{\mathrm{sat}}$ and Π_{gen} below is quoted in units of the intrinsic c_{sat}. The single buffer β_{intra} captures this within-cycle model chemistry. Cross-composition prediction is outside the registered branch; the available NH4Br series is retained only as descriptive supporting-electrolyte context (§2.8).

2.5 The positive-electrode J–Q operating boundary

Because $\Pi_{\mathrm{gen}} \propto i_{\mathrm{app}}$ (the current density J) and $\Pi_{\mathrm{inv}} \propto c_{I_2,\mathrm{tot}}(Q)$ (the areal capacity passed Q), the baseline stress S_{base} is intrinsically a function of the operating point (J, Q),

$$S_{\mathrm{base}}(J,Q) = \gamma_{\mathrm{salt}}\,\frac{\Pi_{\mathrm{inv}}(Q) + \Pi_{\mathrm{gen}}(J)}{\beta_{\mathrm{intra}}(Q)}, \qquad \Pi_{\mathrm{gen}}(J) = \frac{J\,\delta}{2F\,D_{\mathrm{eff}}\,c_{\mathrm{sat}}}. \tag{6}$$

The two thresholds therefore trace two *boundaries* in the (J, Q) plane (Fig. 8). Setting $S_{\mathrm{base}} = 1$ gives the saturation-marker current—the largest J that, at a given Q, keeps free I_2 below saturation:

$$\boxed{J_s^+(Q) = \frac{2F\,D_{\mathrm{eff}}\,c_{\mathrm{sat}}}{\delta}\left[\frac{\beta_{\mathrm{intra}}(Q)}{\gamma_{\mathrm{salt}}} - \Pi_{\mathrm{inv}}(Q)\right]_+ .} \tag{7}$$

The production accessibility relation (§2.6) defines a closure-conditioned boundary $J^{+}_{b,\mathrm{cal}}(Q)$ by $\langle\theta_{\mathrm{cal}}(J,Q)\rangle = \frac{1}{2}$. It lies above J^{+}_{s} over the registered baseline trajectory, separated from it by ΔQ_{cal}. A fixed-current cut returns $Q_{\mathrm{f,cal}}(J)$ and a fixed-capacity cut returns $J^{+}_{b,\mathrm{cal}}(Q)$; both are projections of the calibrated model boundary. The physical-island closure defines a different boundary and, in the matched baseline solve, does not reach half accessibility by $120\,\mathrm{mAh\,cm^{-2}}$. The experimental thickness and rate series therefore provide descriptive stress tests, not calibration or validation of this boundary. Supporting-electrolyte composition is treated as a separate context module in the SI so that NH_4Br does not become the organizing variable of the positive-electrode model. Condition-labelled values of $i^{*} = 2Fk$ are plotted only as external dissolution scales (§2.7); they are not combined with $J^{+}_{b,\mathrm{cal}}$ into a ceiling for this electrolyte.

2.6 Production state markers and physical accessibility comparators

The registered production trajectory uses two state markers:

1. **Baseline averaged-saturation marker**, reached at areal capacity Q_{s} at the registered averaging-operator crossing
$$\langle c_{\mathrm{I_2,surf,free}}/c_{\mathrm{I_2,sat}}\rangle_{\mathrm{aveop1}} = 1.$$
This is the registered positive-electrode averaging-operator crossing, not the first local supersaturation, first auxiliary phase-transfer source or first solid. Spatially local $S > 1$ can activate the native-solid path before the electrode-average crossing. Here $c^{\mathrm{eff}}_{\mathrm{sat}} = c_{\mathrm{sat}}/\gamma_{\mathrm{salt}}$, where $c_{\mathrm{sat}} = 1.33\,\mathrm{mol\,m^{-3}}$ is the dilute-water I_2 solubility reference [31] and $\gamma_{\mathrm{salt}} \approx 3$ is a salting-out factor, a Setschenow-scale reduction of neutral-I_2 solubility at the electrolyte's high ionic strength, carried as a bounded prior rather than a fitted constant and informed only by a cross-salt high-ionic-strength analogue [14] (so $c^{\mathrm{eff}}_{\mathrm{sat}} \approx 0.44\,\mathrm{mol\,m^{-3}}$; SI). This factor sets the *absolute* saturation-marker scale, and the effective solubility is accordingly the second-ranked sensitivity lever (§4.8); the *separation* of the two thresholds, by contrast, is an internal consequence of the selected closures and is compared with distinct retained-versus-blocking physical priors in §4.1.

2. **Production half-accessibility marker**, reached at $Q_{\mathrm{f,cal}}$, where $\langle\theta_{\mathrm{cal}}(\mathbf{x},Q)\rangle = \frac{1}{2}$. In the registered production trajectory, $Q_{\mathrm{f,cal}} = 99.6\,\mathrm{mAh\,cm^{-2}}$ and $\langle\varepsilon_s\rangle = 1.18572\times10^{-4}$ at that event. This trajectory value is not a physical material threshold. The dense isolated-island half-accessibility fraction $\varepsilon_{\mathrm{s,island},1/2} = 1.7973\times10^{-3}$ and the geometric film-percolation marker $\varepsilon^{*}_{\mathrm{s,film\text{-}perc}} = 2.2324\times10^{-3}$ are separate physical comparators.

The production retained-but-not-blocking interval is $\Delta Q_{\mathrm{cal}} = Q_{\mathrm{f,cal}} - Q_{\mathrm{s}}$ and is conditional on the calibrated closure. No single ε^{*}_{s} is used across closure families. Distinct precipitation and clogging stages have physical precedent in porous-media and electrochemical systems [25, 32], while nanoconfined solid iodine can remain electrochemically recoverable [15, 16]; these analogues motivate the separation but do not transfer a numerical threshold to the present felt. The full COMSOL implementation accumulates native deposited solid through the super-saturation-activated Tafel path defined below (Eq. 15). For the analytic operating-map reduction only, we approximate that path by a trajectory-calibrated accumulation law. We track ε_{s} as the deposited-I_2(s) volume fraction on the *total-electrode-volume* basis throughout (consistent with the continuum closure $\varepsilon = \varepsilon_0 - \varepsilon_{\mathrm{s}}$ and the SI), so against the charge passed per electrode area $Q_C = \int i_{\mathrm{app}}\,\mathrm{d}t$,

$$\frac{\mathrm{d}\varepsilon_{\mathrm{s}}}{\mathrm{d}Q_C} = \frac{V_m}{2FL}\,\phi_{\mathrm{ppt}}\,(S-1)_{+}, \qquad (8)$$

where V_m is the molar volume of $I_2(s)$, L the electrode thickness, $(\cdot)_+$ the positive part, and $Q_C = 3.6 \times 10^4 Q$ for Q in $\mathrm{mAh\,cm^{-2}}$. Because ϕ_{ppt} multiplies the super-saturation overshoot $(S-1)_+$, it is a *reduced solid-formation efficiency*, not the native-path current and not a mass-partition fraction of the total generated iodine. Its baseline value and provenance are given in the SI, and its leverage on the production trajectory is swept in §4.5. The physical single-fibre closure uses the pore-volume rescaling $\hat{\varepsilon_{\mathrm{s}}} = \varepsilon_{\mathrm{s}}/\varepsilon$, whereas the production accessibility relation is evaluated on its own regularized solid state. In the reduced bare-path expression, $a_{\mathrm{b}} \simeq a_s(1-\theta)$ with $\theta = \theta_{\mathrm{cal}}$ in the production branch and $\theta = \theta_{\mathrm{island}}$ in the matched physical branch; the full implementation additionally multiplies the bare area by $K_{\mathrm{rel}}^{1/2}$ and retains the separate native-solid path on a_s. A fixed-pore-occupancy scaling $Q \propto \varepsilon L$ is therefore a physical comparator, not the definition of $Q_{\mathrm{f,cal}}$; the existing single-cell thickness series tests only its directional plausibility.

2.7 External dissolution-derived current scales

An areal molar dissolution flux k can be expressed on a two-electron current scale as

$$i^* = nFk, \qquad n = 2, \tag{9}$$

which is the dimensional conversion used by Zhao *et al.* [19]. Their reported values are method- and iodide-concentration-dependent (§4.6); Eq. (9) does not make them a measured current ceiling for the present electrolyte.

The half-cycle distinction is essential. The measurement directly describes solid-iodine redissolution, so it constrains discharge or rest/recovery clearance. Charge oxidizes iodide and does not require solid to dissolve; therefore i^* is not a direct cap on instantaneous charge current. We use it only to motivate a conditional clearance-versus-generation hypothesis: if dissolution or complexation cannot clear oxidized iodine as quickly as it is generated, free-iodine stress and retained inventory can increase. The production saturation marker remains defined by Eq. (5), and the production half-accessibility marker remains defined by $\langle \theta_{\mathrm{cal}} \rangle = 0.5$. The assumed rate laws and condition-labelled literature values are retained as sensitivity context, not as an independently identified failure law.

2.8 Supporting-electrolyte context

The registered positive-electrode model is formulated at fixed $1\,\mathrm{M}$ ZnI_2+$4\,\mathrm{M}$ NH_4Br. NH4Br is a representative supporting electrolyte: bromide can participate in iodine complexation but is not the modeled faradaic protagonist, and ammonium may also affect the negative electrode [33–35]. The available composition series has only 1/1/1/2/1 independent cells across 0/1/2/3/4 M and cannot identify a transferable solubility, complexation or transport coefficient. We therefore do not propagate the previous multi-parameter composition regression into the production boundary. Its deterministic descriptive reconstruction is reported in the SI; prediction at a new supporting electrolyte requires independent speciation, diffusivity, conductivity and viscosity inputs.

3 Continuum implementation and numerical readout

This section is the continuum implementation behind the reduced criterion: a positive-electrode porous-media solve under full-cell boundary conditions that turns the state variables into the spatial fields $S(x,Q)$, $\varepsilon_{\mathrm{s}}(x,Q)$, $\theta(x,Q)$ and the J–Q operating boundary. The scales play distinct, non-interchangeable roles—the continuum solve *generates* the state fields (its solved solid fraction is not itself an accessibility measure, and its raw galvanostatic voltage is not held to experimental accuracy), the molecular and mesoscale layers provide bounded priors or physical comparators, and the experiments have stated descriptive or calibration roles.

A positive-electrode continuum solve under full-cell boundary conditions tests the dimensionless theory and integrates its closures. The modeled domain is the positive carbon felt (Fig. 3): a 2 mm-thick porous slab resolved in two dimensions, with the through-plane coordinate x running from the current-collector face to the separator/membrane face (which faces the zinc negative) and the in-plane coordinate y (0–20 mm) along the electrolyte flow; the reaction current and modeled solid inventory are nonuniform across the through-plane domain. These are continuum fields, not a resolved deposit geometry. A tertiary-current-distribution model [20–23] solves the governing balances of §3.1 (Eqs. (11)–(16)) in the real cell geometry, with two-way Darcy/porous-media flow: the velocity is solved and coupled back into transport, and the permeability feedback stays mild within the analyzed window (§4.7). The solve includes a constant-exchange-current Butler–Volmer soluble-iodine path, an empirical oxidized-pool Nernst-like equilibrium potential, and a local-supersaturation-activated native-solid Tafel path for the same overall $2\,\mathrm{I}^- \to \mathrm{I}_2 + 2e^-$ couple. The native-solid path, not the parallel auxiliary phase-transfer tracker, advances the deposited-species volume used by the accessibility, permeability and transport closures of §2.2. The reported model terminal voltage is the solved galvanostatic scaffold response, not an empirical fit or an independently validated full-cell prediction; numerical details are in the Methods. For *interpreting* the voltage we additionally use a Butler–Volmer reduction,

$$V(Q) = V_0 + \underbrace{\frac{RT}{2F}\ln\left(\frac{c^{\mathrm{f}}_{\mathrm{I}_2}}{c^{\circ}_{\mathrm{obs}}}\right)}_{\text{concentration term}} + b_T\left[\underbrace{\ln\frac{c^0_{\mathrm{I}^-}}{c_{\mathrm{I}^-}(Q)}}_{\text{iodide depletion}} + \underbrace{\ln\frac{1}{A_{\mathrm{bare}}(Q)}}_{\text{accessibility}}\right] + i_{\mathrm{app}}R_\Omega, \tag{10}$$

where $c^{\circ}_{\mathrm{obs}} = 1\,\mathrm{mol\,m^{-3}}$ makes the logarithm dimensionless (a different reference is absorbed into V_0). Here V_0 and the apparent Tafel-like coefficient b_T are fitted observation-layer coefficients, not independently identified physical kinetic parameters.

3.1 Governing balances of the positive-electrode scaffold

The reduced criterion of §2 distills a continuum iodine-state scaffold; we state its minimal closed set here so that the baseline stress S_{base} reads as a consequence of an explicit porous-electrode balance rather than an assertion. The full implementation (advective transport, reservoir and boundary conditions, parameter provenance) is given in the SI. The solved pore-phase species is the lumped oxidized pool c_O (internal field `cBr2`); the free-iodide coordinate is an algebraic inventory,

$$c_R = \max(c_{R,\mathrm{floor}}, c_{R,0} - 3c_O)\,, \tag{11}$$

$$\varepsilon_0\,\partial_t c_O + \nabla\cdot(\mathbf{u}c_O - D_{O,\mathrm{eff}}\nabla c_O) = R_{\mathrm{b}} - R_{\mathrm{dep}} + \frac{r_{\mathrm{d}} - r_{\mathrm{p}}}{V_m}. \tag{12}$$

Here $R_{\mathrm{b}} = a_{\mathrm{b}}j_{\mathrm{b}}/(2F)$ is soluble-iodine production on accessible carbon, $R_{\mathrm{dep}} = a_s j_{\mathrm{dep}}/(2F)$ is conversion into the native deposited species, and $r_{\mathrm{p}}, r_{\mathrm{d}}$ are the much smaller auxiliary volume-fraction rates defined below. The factor three in Eq. (11) is the registered stoichiometric inventory closure for oxidation plus triiodide binding; it is not a second transported-species equation. Charge is conserved through the sum of the two positive-electrode paths,

$$\begin{aligned} &i_v = a_{\mathrm{b}}j_{\mathrm{b}} + a_s j_{\mathrm{dep}}, && \\ &\nabla\cdot\mathbf{i}_s = -i_v, && \mathbf{i}_s = -\sigma_{s,\mathrm{eff}}\nabla\phi_s, \\ &\nabla\cdot\mathbf{i}_l = +i_v, && \mathbf{i}_l = -\kappa_{l,\mathrm{eff}}\nabla\phi_l. \end{aligned} \tag{13}$$

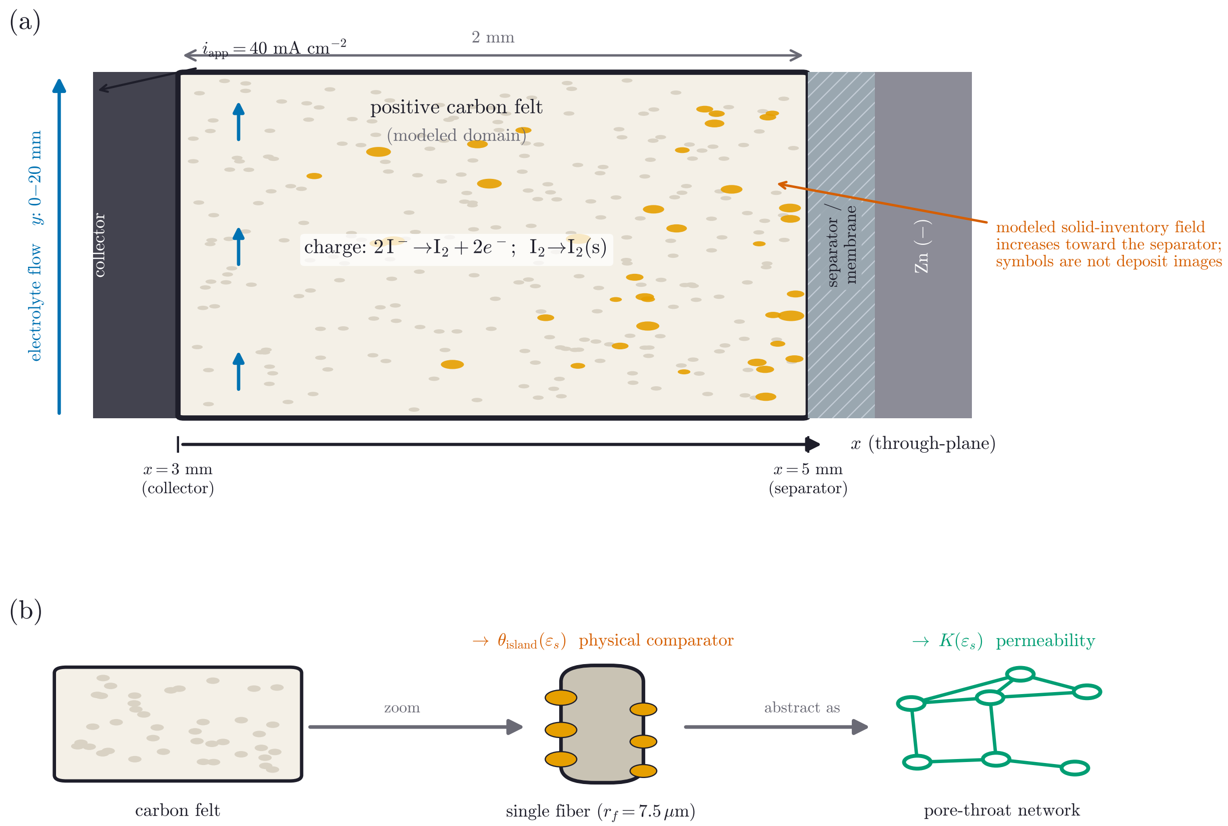

Figure 3: Modeled system and the positive-electrode domain. (a) The continuum model resolves the positive carbon felt in two dimensions: a 2 mm-thick porous slab between the current collector (through-plane coordinate $x = 3$ mm face) and the separator/membrane ($x = 5$ mm face, which faces the zinc negative), with electrolyte flowing along the in-plane coordinate y (0–20 mm). Charge oxidizes iodide, $2\,I^- \to I_2 + 2e^-$; local super-saturation activates the native deposited-species path. The schematic distinguishes soluble production, native-solid formation and accessibility loss rather than asserting a resolved morphology. Orange symbols schematize retained inventory and are not deposit images or a morphology assignment. The solve returns the through-plane fields S, ε_s, the selected θ, the flow velocity and the current; the maps of Fig. 4 and Supplementary Fig. S8 are slices of this same domain (Fig. 4 averaged over y). (b) The mesoscale models provide physical comparators: a single-fiber unit cell ($r_f = 7.5\,\mu$m) supplies the isolated-island family $\theta_{island}(\varepsilon_s, N)$, and a pore-throat network supplies the smooth-permeability response $K(\varepsilon_s)$. Neither fixes the voltage-calibrated production magnitude.

The bare-carbon path uses a constant prescribed exchange-current density; concentration enters through the implemented equilibrium-potential expression rather than through reaction orders,

$$\begin{aligned}
j_{\mathrm{b}} &= i_0\left[e^{\alpha_a F\eta/RT} - e^{-\alpha_c F\eta/RT}\right],\\
\eta &= \phi_s - \phi_l - E_{\mathrm{eq}},\\
E_{\mathrm{eq}} &= E^{0\prime} + \gamma_a \frac{RT}{2F}\ln\left[\frac{\max(c^{\mathrm{f}}_{\mathrm{I}_2}, c_{\mathrm{f}})}{c_{\mathrm{ref}}/\beta}\right] + \eta_{\mathrm{tank}},\\
a_{\mathrm{b}} &= a_s(1-\theta)K_{\mathrm{rel}}^{1/2}.
\end{aligned} \tag{14}$$

The registered values are $i_0 = 5\times 10^4\ \mathrm{A\,m^{-2}}$, $\alpha_a = \alpha_c = 1$, $E^{0\prime} = 0.548\,\mathrm{V}$, $c_{\mathrm{ref}} = 120\,\mathrm{mol\,m^{-3}}$, $c_{\mathrm{f}} = 0.02\,\mathrm{mol\,m^{-3}}$ and $\gamma_a = 1$. Because $c^{\mathrm{f}}_{\mathrm{I}_2} = c_O/\beta$, the β terms cancel away from the numerical floor; Eq. (14) is therefore an empirical total-oxidized-pool Nernst-like potential, not a full activity model for the $\mathrm{I_2/I^-}$ couple. The tank correction switches on only after the registered charge window and is zero for the results reported here.

The native-solid path uses COMSOL's anodic Tafel form and is activated by the *local* free-iodine stress,

$$\begin{aligned}
&j_{\mathrm{dep}} = i_{0,\mathrm{dep}}\,10^{\eta/A_a}, \qquad i_{0,\mathrm{dep}} = i^{\mathrm{base}}_{0,\mathrm{dep}}\max(S_{\mathrm{local}}-1, 0),\\
&\partial_t\varepsilon_{\mathrm{s}} = V_m R_{\mathrm{dep}}, \qquad \partial_t\varepsilon_{\mathrm{s,aux}} = r_{\mathrm{p}} - r_{\mathrm{d}}, \qquad \partial_t N_{\mathrm{I}_2} = \kappa_N(N_{\mathrm{target}} - N_{\mathrm{I}_2}).
\end{aligned} \tag{15}$$

Here $i^{\mathrm{base}}_{0,\mathrm{dep}} = 0.05\,\mathrm{A\,m^{-2}}$ and $A_a = 118\,\mathrm{mV}$. The deposited species ε_{s} in Eq. (15) is the inventory used by every transport and accessibility closure. The parallel smooth phase-transfer state $\varepsilon_{\mathrm{s,aux}}$ reaches 1.08×10^{-6} at the endpoint and enters the c_O balance only through $(r_{\mathrm{d}} - r_{\mathrm{p}})/V_m$; it does not drive N_{I_2}. The latter evolves in parallel by its own target-relaxation equation and enters the calibrated accessibility relation. By comparison, $\varepsilon_{\mathrm{s}} = 3.219\times 10^{-3}$ for the native deposited species. The deposited path carries 0.552% of the integrated charge but 28.0% of the endpoint positive current. Local activation before the electrode-average $S = 1$ crossing makes Q_{s} an averaged saturation coordinate rather than a first-solid event. The solid feeds back on transport only through algebraic closures,

$$\begin{aligned}
&\varepsilon = \varepsilon_0 - \varepsilon_{\mathrm{s}}, \qquad \{D, \kappa_l\} \propto \left(\tfrac{\varepsilon}{\varepsilon_0}\right)^{3/2},\\
&K = \left(\tfrac{\varepsilon}{\varepsilon_0}\right)^{3}\left(\tfrac{1-\varepsilon_0}{1-\varepsilon}\right)^{2}, \qquad \theta = \mathcal{C}_b(\varepsilon_{\mathrm{s}}),
\end{aligned} \tag{16}$$

with ε_0 the felt porosity, ε the running pore-liquid fraction (Table 1), K the relative permeability, and $\mathcal{C}_b$ the registered branch-specific accessibility relation: θ_{cal} for production or θ_{island} for the dense physical comparator. The headline relation $D_{\mathrm{eff}} = D_0\varepsilon^{1.5}$ thus carries the full Bruggeman modifier $\varepsilon^{1.5}$ on the free diffusivity (the baseline Π_{gen} below is quoted at $D_{\mathrm{eff}}\to D_0$). That modifier factors as $\varepsilon^{1.5} = \varepsilon_0^{1.5}(\varepsilon/\varepsilon_0)^{3/2}$ into a felt-porosity part $\varepsilon_0^{1.5}$ that responds to *compression* and a residual $(\varepsilon/\varepsilon_0)^{3/2}$ that responds to native-solid *loading*.

Of these, $c_O, \phi_s, \phi_l, \varepsilon_{\mathrm{s}}, \varepsilon_{\mathrm{s,aux}}$ and N_{I_2} are solved states; $c_R, \varepsilon, D, \kappa_l, K$ and θ are algebraic consequences or closures evaluated on them; and the super-saturation S with the dimensionless groups of §2.3 is the reduced coordinate read off the solved state. This separation is made explicit in the model stack and the implementation-to-theory map of the SI. Convection enters as a solved Darcy/porous-media field whose velocity is computed and fed back into transport (full two-way coupling). In practice this feedback is negligible because native-solid loading barely lowers the permeability before failure (§4.7): the solved velocity varies by $< 0.1\%$ across a charge, and a frozen-velocity control gives the same saturation marker within the reported numerical tolerance. Finally, integrating both faradaic paths across the thickness gives $\frac{1}{2F}\int_0^L (a_{\mathrm{b}} j_{\mathrm{b}} + a_s j_{\mathrm{dep}})\,\mathrm{d}x = i_{\mathrm{app}}/2F$. The reduced

Π_{gen} uses this total charge-equivalent supply and carries no explicit $a_s L$ factor; its interpretation as soluble-pool generation is an approximation bounded by the 0.552% integrated native-solid-path share.

The model's assumption-to-claim boundaries are consolidated in Supplementary Table S15. The release-critical distinctions are that the positive electrode is solved while the zinc electrode is not; speciation and the Nernst layer are reduced closures; ε_{s} is an inventory rather than an accessibility measurement; production θ_{cal} is calibrated; physical θ_{island} is a separate morphology-prior family; and $i^* = 2Fk$ is a condition-labelled external scale rather than a present-cell ceiling.

Taken together, these scales form an evidence map rather than a parameter-free chain: each calculation has a registered role in the reduced continuum criterion or its physical bounds (Fig. 2). This makes explicit what each scale is *for*: DFT supplies only a relative placement tendency, not the island density N; MD supplies a bounded oxidized-carrier diffusivity prior; and the single-fiber and pore-network calculations bound physical accessibility and smooth-permeability responses. The voltage-calibrated production relation remains a separate model object. Its implementation-to-theory map is tabulated in the SI.

4 Results

The results are organized around one question—*when does soluble or retained iodine become associated with modeled loss of positive-electrode accessibility?* We first resolve a registered production charge (§4.1), then isolate the information added by the accessibility closure (§4.2). Molecular calculations provide placement and diffusivity priors (§4.3); single-fiber and pore-network calculations provide distinct physical bounds (§4.4); matched continuum solves quantify model-form sensitivity without assigning a unique deposit morphology. We then map the conditional operating envelope (§4.5), compare it with condition-labelled dissolution kinetics (§4.6), bound alternative transport channels (§4.7), and rank model-internal levers (Table 3, §4.8).

What the model does and does not predict. Four results are carried forward with explicit evidence labels. (i) Within the registered production branch, the registered averaged-saturation marker precedes the voltage-calibrated half-accessibility marker. (ii) The resulting interval ΔQ_{cal} is a closure-conditioned model output, not a universal reserve. (iii) Published iodine-dissolution measurements provide an external, method- and concentration-dependent kinetic scale; they do not independently determine the sustainable current of the present electrolyte. (iv) The sensitivity analysis identifies model-internal directions and dominant priors, while absolute capacities remain conditional on c_{sat}, D_{eff}, γ_{salt} and the accessibility closure. Cycle-level reset and deposit identity are outside the solved single charge (§5.1).

4.1 The registered production charge: saturation, retention, and calibrated accessibility

The load-bearing production-model result is the separation between the saturation marker $Q_{\text{s}} = 83.0\,\text{mAh}\,\text{cm}^{-2}$ and the calibrated half-accessibility marker $Q_{\text{f,cal}} = 99.6\,\text{mAh}\,\text{cm}^{-2}$, giving $\Delta Q_{\text{cal}} = 16.6\,\text{mAh}\,\text{cm}^{-2}$. These are events on the registered `PB-R526-J40-Q120-DSET5-v1` trajectory and are not physical-island thresholds or experimental landmarks. The production solve separates quantities that an iodine-inventory metric would merge: averaged free-I_2 saturation, retained solid, and the response of the calibrated accessibility state (Table 2, Fig. 4). The positive-electrode `aveop1` stress crosses $S = 1$ at Q_{s}, about 40% positive-electrode state of charge. Native-solid-path current is non-zero at every resolved node by $Q = 96$, where the volume-averaged ε_s is 7.13×10^{-5}; these are solver readouts, not experimental solid-formation detections. The correspondingly averaged production response reaches $\langle\theta_{cal}\rangle = 0.5$ and 0.9 at $Q = 99.6$ and $111.2\,\mathrm{mAh\,cm^{-2}}$, respectively, and its modeled $\mathrm{d}V/\mathrm{d}Q$ maximum follows at $116.7\,\mathrm{mAh\,cm^{-2}}$. One half is a declared midpoint of the calibrated accessibility response, not a universal percolation threshold or an identified morphological transition. The alternative 0.01 and 0.9 criteria are reported only to show criterion sensitivity. Quantitatively, the production generation group is $\Pi_{gen} = 77.9$ (at the free-diffusivity reference so that compression enters as the $D_{eff} = D_0\varepsilon^{1.5}$ modifier). At $Q = 0$ the stored oxidized inventory is negligible, so the initial surface total (78.0 in units of c_{sat}) *is* this generation term rather than pre-existing iodine. Over the charge the total oxidized-iodine inventory rises 4.16× while the *free*, precipitable fraction rises 9.48×, because iodide is consumed 58% and the buffer collapses from $\beta = 1263$ to 554. In aqueous polybromide catholytes, separate-phase or vapor bromine emergence is state-of-charge- and complexant-dependent [36]. In a distinct static Zn–I_2/KB cell, in-situ UV–visible spectroscopy shows the dissolved I_3^- concentration rising sharply near 1.45 V [37]. These observations are chemistry-specific analogues, not confirmation of free-halogen emergence in the present ZIFB. The as-solved two-dimensional field maps (Supplementary Fig. S8) and the through-plane space-time evolution (Fig. 4) make this concrete: solid I_2 and the total current localize in a thin band at one electrode face while the free-I_2 stress remains a smooth gradient, and the calibrated accessibility-loss region builds at that face as capacity increases. That the reduced stress S_{base} reads back the solver's exported super-saturation channel up to a single known normalization (slope 3.00, from the free-monomer reference $c_{sat}/3$) is an algebraic consistency check (SI), not an independent validation.

Table 2: Registered production-state events and criterion sensitivity. Model outputs for the registered $J = 40\,\mathrm{mA\,cm^{-2}}$ production charge. The θ_{cal} criteria are readouts of the calibrated production relation and are not physical-island or experimental thresholds.

event	criterion	Q ($\mathrm{mAh\,cm^{-2}}$)	meaning
averaged-saturation marker	$\langle S\rangle_{aveop1} = 1$	83	registered mean free-I_2 stress reaches unity; not first local solid
first production-accessibility response	$\langle\theta_{cal}\rangle = 0.01$ (criterion-sensitivity entry)	85	earliest response under the calibrated closure, near the resolution floor
native solid finitely resolved	solid-path current at every node; averaged $\varepsilon_s = 7.13 \times 10^{-5}$	96	retained $I_2(s)$ registered throughout the positive domain
production half-accessibility marker	$\langle\theta_{cal}\rangle = \frac{1}{2}$ ($\equiv Q_{f,cal}$)	99.6	declared midpoint of the calibrated response
production high-accessibility-loss marker	$\langle\theta_{cal}\rangle = 0.9$	111.2	criterion-sensitivity readout
voltage symptom	$\mathrm{d}V/\mathrm{d}Q$ peak	116.7	downstream observable

The native-solid path and the surviving bare-carbon path retain partial spatial overlap. We re-exported the true partial currents directly from the registered `dset5`→`sol5` branch at six capacities and 5995 positive-domain nodes per snapshot. The exported arealized partial currents are $J^A_b = W_{cf} a_b j_{b,loc}$ for the bare-carbon path and $J^A_{dep} = W_{cf} a_s j_{dep,loc}$ for the native-solid path,

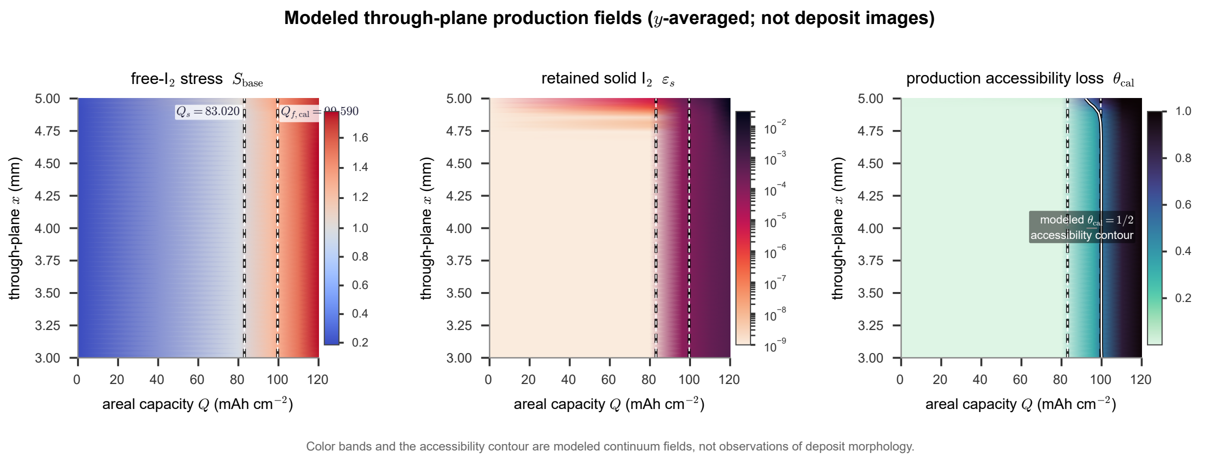

Figure 4: Through-plane evolution of the registered production state. Each panel is a field from the 2D domain of Fig. 3, averaged over the flow direction y and plotted against through-plane position x (vertical; collector at 3 mm, separator at 5 mm) and areal capacity passed Q (horizontal, the charge "clock"). Left to right: free-I_2 stress S, retained solid fraction ε_s, and calibrated accessibility loss θ_{cal}. The dashed verticals mark the production saturation marker $Q_s = 83.0$ and calibrated half-accessibility marker $Q_{f,cal} = 99.6$ mAh cm^{-2} (Table 2). The stress rises smoothly; retained solid is concentrated near the separator-facing side; and the production-closure half-accessibility contour develops there first. The contour is a solved state of the calibrated closure, not an observed blocking front or an identified deposit morphology.

where W_{cf} is the 2 mm felt thickness. The latter is non-zero at every exported node by $Q = 96$ mAh cm^{-2}. At $Q = 120$, define the "solid-rich half" as the unweighted exported nodes above the within-snapshot median ε_s. On that descriptive point-sampling basis, 98.6% of the native-solid-current magnitude and 84.3% of the bare-path-current magnitude fall in that half. The closure-defined bare area is, by construction, anti-correlated with retained solid (Spearman $\rho = -0.99$), while the intensive bare-carbon rate (among the 5615 nodes with $a_b > 0$) and native-solid current have positive rank correlations with solid ($\rho = +0.96$ and $+0.96$); the extensive bare-path current has only a weak positive rank correlation ($\rho = +0.20$). Its raw Pearson coefficient has the opposite sign ($r = -0.21$). Because these exported points carry no quadrature weights and the summaries are metric-dependent, neither sign is used to infer general exclusion or a morphology. A separate positive-electrode averaging-operator ledger gives a native-solid-path share of 28.0% at the endpoint but only 0.552% integrated over the full charge. Thus the spatial table is a descriptive readout of the imposed accessibility closure, not independent validation of blocking. The frozen node table and analysis are included in the release source data.

4.2 An accessibility closure adds information and changes the coupled trajectory

The ablation is a bookkeeping test on the production state (Fig. 5). The stress S identifies the modeled saturation crossing but contains no accessibility state; ε_s records retained inventory but contains no declared functional criterion. The calibrated relation $\theta_{cal}(\varepsilon_s)$ maps that inventory to the production accessibility response, including $Q_{f,cal}$. Thus S, ε_s, θ_{cal} and $A_{bare} = 1 - \theta_{cal}$ carry distinct information on one capacity axis. This does not make $\langle\theta_{cal}\rangle = 0.5$ a physical percolation threshold, nor does it identify N: the physical isolated-island response and its half-accessibility fraction are analyzed separately. The same inventory–current decoupling is observed directly across published cells: at fixed iodine inventory the deliverable capacity collapses about five-fold when the current density is raised five-fold and recovers fully on returning to the low rate [17]. In one biphasic Zn–I system, pushing iodine concentration from the robust 1.7–2.2 M range to 2.7 M worsened rate stability [38]; reducing loading extended high-rate cycling in a separate design [39]. Together these reports show that stored amount alone does not fix high-rate utilization; they do not establish a universal current ceiling for the present electrode. The next two subsections provide bounded molecular and mesoscale priors for D_{eff}, physical island accessibility and $K(\varepsilon_{\mathrm{s}})$; they do not bottom-up determine the calibrated production magnitude.

4.3 Molecular priors: where iodine is retained and how fast it clears

The chain begins at the molecular scale, which supplies *bounds and tendencies*—not rate constants—for two quantities the upper layers inherit. Periodic CP2K places molecular I_2 on the carbon surface with single-I_2 adsorption energies of −0.40 (basal), −0.85 (carbonyl, C=O) and −1.00 eV (hydroxyl, C−OH), and −0.32 eV at a vacancy. Within the registered finite-cell setup, the more negative oxygen-site values establish only a relative placement ordering. A compact two-I_2 configuration at C−OH is lower in electronic energy by 0.69 eV than the registered separated reference (Fig. 6a; charge-density-difference maps and projected densities of states in the SI). Together with independent DFT of iodine on carbon [40, 41], these calculations motivate testing relative siting and local co-association tendencies only; they do not provide solution free energies, trapping events, a nucleation pathway, island density N, a growth law, or the production accessibility relation. Accordingly, N is swept as a morphology prior in the physical single-fiber calculations rather than inferred from DFT. A single-point cluster diagnostic gives electronic-energy changes of −0.46 eV for the $I^-{+}I_2$ cluster and +0.05 eV for the $Br^-{+}I_2$ cluster (Fig. 6b). These are neither solution free energies nor equilibrium constants and do not set K_{I} or K_{Br}; the numerical constants remain conditional literature priors [14, 42].

Classical molecular dynamics supplies a bulk-electrolyte mobility prior. Under the ECC-scaled parameterization, the diffusivities of I^-, $I_3{}^-$ and I_2Br^- are 1.22, 0.75 and 0.72 × 10^{-9} $\mathrm{m^2\,s^{-1}}$, respectively (Fig. 6c). Thus the oxidized carriers are 38–41% slower than iodide within this force-field set. The continuum value $D_{\mathrm{eff}} = 0.5 \times 10^{-9}\ \mathrm{m^2\,s^{-1}}$ lies inside the bracket formed by the ECC and formal-charge runs. Each SOC/charge case contains one 20 ns production trajectory. The error bars aggregate four contiguous time-block stability estimates over the five distinct SOC boxes; they are not independent-replica standard errors. This is a bounded bulk-electrolyte prior: the MD does not simulate transport through deposited iodine, validate viscosity or determine an accessibility closure.

4.4 Physical mesoscale bounds and matched closure sensitivity

The single-fiber model supplies a physical isolated-island family, not the magnitude of the production relation (the geometry is rendered in Supplementary Fig. S9a). For the dense fitted case,

$$\theta_{\mathrm{island}} = 1 - \exp\left[-35.4\,\varepsilon_{\mathrm{s,reg}}^{0.6222}\right], \tag{17}$$

and its half-accessibility fraction is $\varepsilon_{\mathrm{s,island},1/2} = 1.7973 \times 10^{-3}$. Evaluated one way on the production $\varepsilon_{\mathrm{s}}(Q)$ trajectory, the sparse, DFT-clustered, moderate and dense cases end at $\theta_{\mathrm{island}} = 0.116$, 0.329, 0.404 and 0.631, respectively; the dense shadow projection crosses one half only at $Q = 118.64\ \mathrm{mAh\,cm^{-2}}$. These curves are morphology-prior sweeps with no feedback, and DFT does not set their N values.

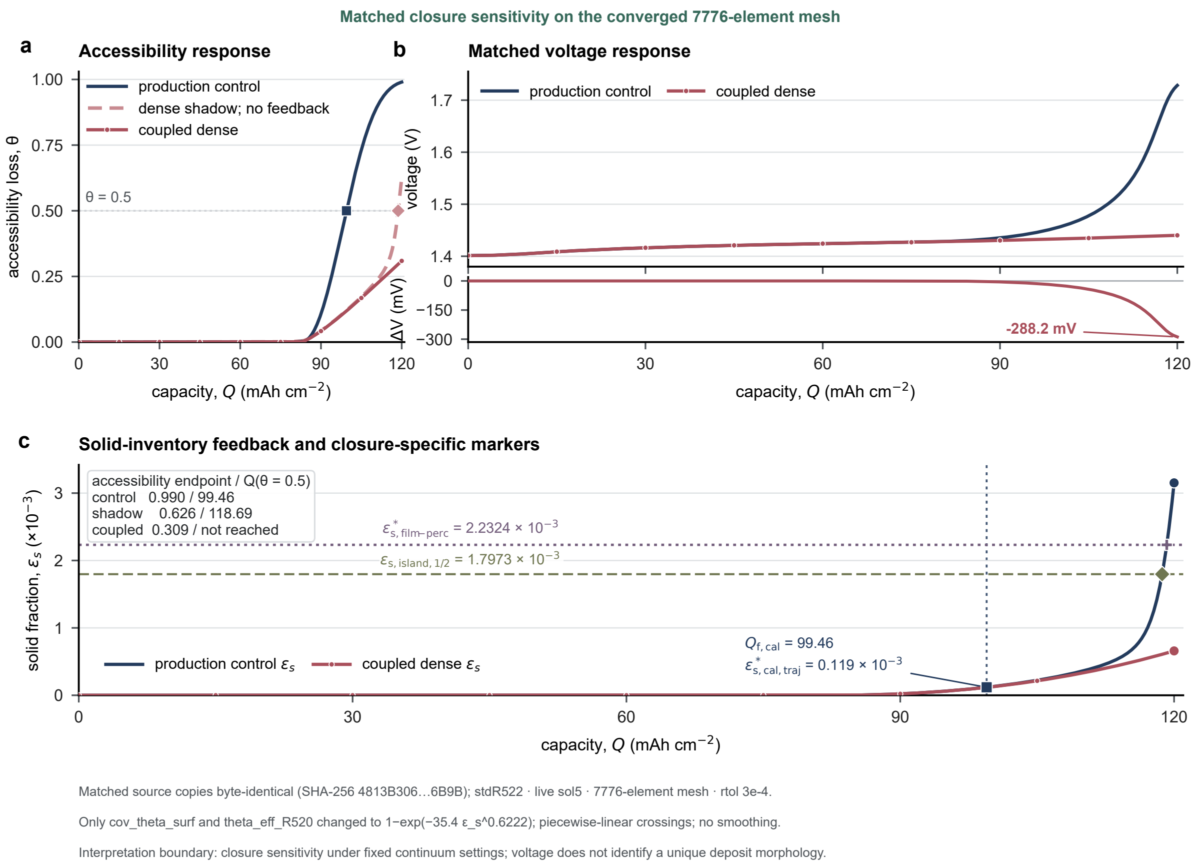

Figure 5: Matched true-mesh accessibility-closure sensitivity. (a) Accessibility loss for the production control, a one-way dense-island shadow evaluated on the control inventory (no feedback), and the coupled dense-island solve. (b) Matched terminal-voltage trajectories and their physical-minus-control difference. (c) Each coupled branch's own retained-solid trajectory, with the production-trajectory readout, dense-island half-accessibility fraction and separate film-percolation comparator kept distinct. The two 7776-element solves used byte-identical source copies, the same study, live solution, time grid and rtol $= 3 \times 10^{-4}$; only the two registered accessibility expressions changed. Both the tighter-tolerance and mesh gates passed. The endpoint difference is $-288.2\,\mathrm{mV}$, quantifying model-closure sensitivity rather than identifying a real deposit morphology.

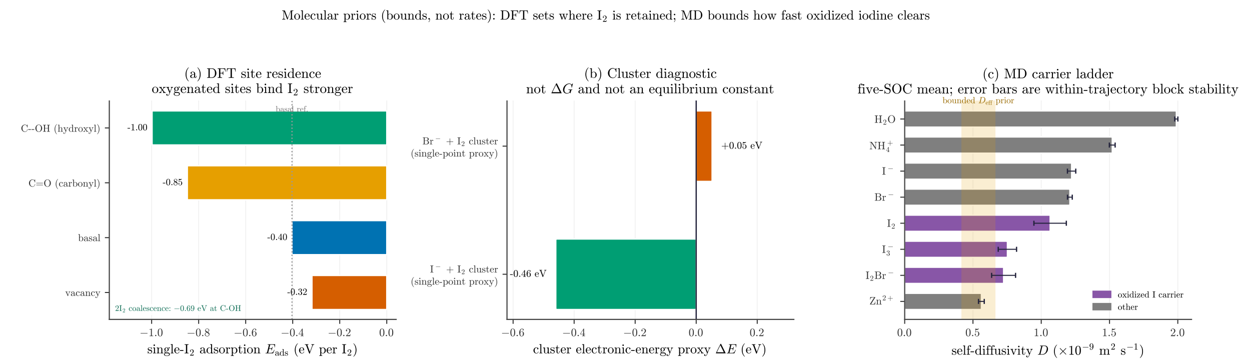

Figure 6: Molecular priors (bounds, not rates). (a) Registered DFT electronic energies establish a relative single-I_2 site ordering (basal -0.40, carbonyl -0.85, hydroxyl -1.00, vacancy -0.32 eV) and place a compact $2I_2$ configuration at C−OH 0.69 eV below the registered separated reference. These finite-cell values are qualitative placement diagnostics, not solution free energies or nucleation evidence. (b) Single-point cluster electronic-energy diagnostic: values are not ΔG and do not determine solution equilibrium constants. (c) MD carrier diffusivities: the oxidized carriers I_3^- and I_2Br^- are 38–41% slower than I^- (neutral I_2 only $\sim$12%); shaded band is the low-end D_{eff} prior. Error bars are propagated four-block within-trajectory stability statistics, not independent-replica SEM.

We then performed a fresh matched true-mesh continuum comparison from byte-identical copies of the canonical input. Both branches use the same study (`stdR522`), live solution (`sol5`), 7776-element mesh, rtol $= 3 \times 10^{-4}$, output grid and all non-accessibility parameters. The control gives $Q_{\text{s}} = 82.890$ and $Q_{\text{f,cal}} = 99.464$ mAh cm^{-2}. The coupled dense-island case gives $Q_{\text{s}} = 82.892$ mAh cm^{-2}, but does not reach $\langle\theta_{\text{island}}\rangle = 0.5$ by $Q = 120$ mAh cm^{-2}. The endpoint states are

$$(\langle\varepsilon_{\text{s}}\rangle, \langle\theta\rangle)_{\text{island}} = (6.586 \times 10^{-4}, 0.309), \qquad (\langle\varepsilon_{\text{s}}\rangle, \langle\theta\rangle)_{\text{cal}} = (3.152 \times 10^{-3}, 0.990).$$

The dense-island terminal voltage is 288.156 mV lower. The true-mesh one-way shadow ends at 0.626 because it uses the control inventory, whereas the coupled case ends at 0.309 because accessibility feedback changes the inventory trajectory. Relative to the same-tolerance 1944-element pair, the endpoint voltage difference changes by only 0.094 mV; all predeclared tolerance and mesh gates pass. This registered comparison quantifies closure sensitivity; it neither validates the production relation nor identifies an island, film or pore-spanning morphology. The production-state plots and the J–Q map retain the canonical registered values ($Q_{\text{s}} = 83.020$ and $Q_{\text{f},cal} = 99.590$ mAh cm^{-2}); the corresponding true-mesh control values differ by only 0.130 and 0.126 mAh cm^{-2}, respectively. Thus, the true-mesh pair is the authoritative closure-comparison branch, while the frozen canonical branch remains the provenance anchor for the registered production-state figures.

Three solid-fraction markers are therefore kept separate. The true-mesh production control has $\langle\varepsilon_{\text{s}}\rangle = 1.190 \times 10^{-4}$ at $Q_{\text{f,cal}}$; the dense physical-island relation reaches one half at 1.7973×10^{-3}; and the separate geometric film-percolation calculation gives $\varepsilon^*_{\text{s,film-perc}} = 2.2324 \times 10^{-3}$. None is substituted for another.

A pore-throat network (Supplementary Fig. S9b) then tests whether deposited I_2 throttles transport by clogging. For *every* one of the six deposit-placement laws swept, the network stays 98–99% permeable at the operating load ($\varepsilon_{\text{s}} \sim 2$–$4 \times 10^{-3}$) and the active backbone fully percolates (Fig. 7b). Halving the permeability requires 30–100× the operating load, so a *smooth geometric* permeability collapse does not drive the calibrated voltage response within the registered smooth-permeability channel. This rules out only that continuum-permeability channel: it does not exclude the free-I_2 mass-transfer stress carried in $\Pi_{\text{gen}} \propto 1/D_{\text{eff}}$, viscosity-driven pressure ($\Delta p \propto \mu/K$), or discrete

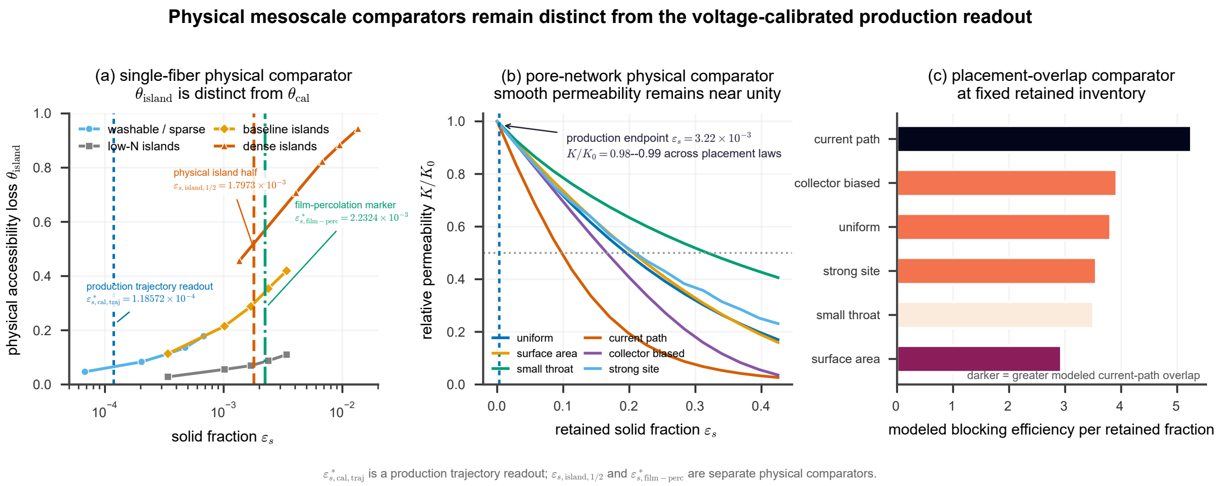

Figure 7: Physical bounds from the mesoscale models. (a) Physical isolated-island accessibility $\theta_{\rm island}(\varepsilon_{\rm s})$ for the tested morphology families; the dense fitted half-accessibility fraction is $\varepsilon_{\rm s,island,1/2} = 1.7973\times10^{-3}$. (b) Pore-network smooth relative permeability remains near unity over the modeled operating load and halves only at much larger solid fractions. (c) At fixed solid inventory, blocking efficiency depends on deposit placement. The separate geometric film-percolation marker is $\varepsilon^*_{\rm s,film\text{-}perc} = 2.2324\times10^{-3}$. These physical bounds do not determine the production voltage-calibrated accessibility magnitude.

local pore-throat/shell blockage, which the network does not resolve (Supplementary Table S14). The network does supply one non-continuum constraint a Kozeny–Carman law misses: blocking efficiency is set by the spatial *overlap* of deposit, not its amount (path-correlated deposition blocks $\sim1.8\times$ more efficiently than surface-spread at the same loading; Fig. 7c). That deposit placement can matter at fixed inventory is consistent with external observations: in an analogous flow cell a coalesced passivating layer collapses efficiency where finely dispersed particles leave the electrode fresh [43], and the accessibility of a retained halogen phase is governed by its physical state [13]. In a non-iodine TEMPO/KBF_4 flow-cell analogue, operando neutron imaging linked unintended salt precipitation to obstructed ion transport and reduced available surface area [44]. The present model keeps these physical bounds separate from the voltage-calibrated production relation.

How each closure enters the model also tells us what it is sensitive to. The selected accessibility relation feeds the continuum kinetics through the accessible-area prefactor $(1-\theta)$, and hence the charge-voltage response, whereas the relative permeability $K(\varepsilon_{\rm s})$ feeds only the Darcy flow. Because K is a *geometric* closure, these two transport channels respond to different electrolyte properties and do not mix: the free-I_2 generation load scales as $\Pi_{\rm gen} \propto 1/D_{\rm eff}$, so it tracks the oxidized-carrier diffusivity, while the hydraulic pressure across the felt scales as $\Delta p \propto \mu/K$, so it tracks the viscosity. A more viscous electrolyte therefore raises pumping pressure without directly moving the averaged-saturation marker, whereas a slower-diffusing one raises $\Pi_{\rm gen}$ and brings that marker forward (Supplementary Table S14). The modular closures make this separation explicit, and it gives a change of supporting electrolyte a quantitative route into the model. The pore network therefore bounds only the *smooth geometric* permeability channel of panel (b); the mass-transfer and viscosity channels of panels (c,d) are separate and remain active, which is why a change of electrolyte that alters diffusivity or viscosity moves the operating point even when the geometric $K(\varepsilon_{\rm s})$ barely does.

4.5 The transport and formulation envelope

The operating boundary is controlled by a small set of transport and formulation levers, and we now map how much each one moves it. Two of them, the intrinsic dilute-water reference c_{sat} (and therefore $c_{\mathrm{sat}}^{\mathrm{eff}}$ at fixed γ_{salt}) and the rate-closure efficiency ϕ_{ppt}, enter the reduced criterion algebraically rather than by re-solving the transport. Their leverage is therefore swept as a postprocess on the baseline trajectory, exact with respect to the reduced criterion, though not a re-solved full cell. Raising c_{sat} delays the averaged-saturation marker strongly: it moves from $Q_{\mathrm{s}} = 65$ to $109\,\mathrm{mAh\,cm^{-2}}$ as c_{sat} rises from 1.0 to 2.0 $\mathrm{mol\,m^{-3}}$. At $c_{\mathrm{sat}} = 2\,\mathrm{mol\,m^{-3}}$ the marker reaches the window edge and the production half-accessibility marker is not reached; above it the registered averaged saturation crossing is absent ($S_{\max} < 1$). Propagated as an uncertainty rather than a lever, the literature spread of the aqueous solubility maps the baseline marker onto an explicit error band for the least-constrained constant: $Q_{\mathrm{s}} \approx 71$ at 1.1 mM, ≈ 76 at 1.2 mM, and 83 at the adopted 1.33 mM. The rate-closure efficiency ϕ_{ppt} sets the width of the modeled interval: lowering ϕ_{ppt} leaves the saturation marker Q_{s} fixed but pushes $Q_{\mathrm{f,cal}}$ outward, widening ΔQ_{cal} from 17 to $29\,\mathrm{mAh\,cm^{-2}}$. The diffusivity channel is a seven-point *solved* COMSOL sweep: the saturation marker rises smoothly from $Q_{\mathrm{s}} = 45$ to $106\,\mathrm{mAh\,cm^{-2}}$ as D_{eff}/D_0 goes from 0.5 to 2, following $\Pi_{\mathrm{gen}} \propto 1/D_{\mathrm{eff}}$. Because the mass-transfer stress depends only on the combination δ/D_{eff}, the boundary-layer thickness and the diffusivity collapse onto one axis: both production markers slide together as δ/D_{eff} varies. The free-I_2 generation stress, not c_{sat} alone, therefore remains central to this marker. The hydraulic consequence is separate again: viscosity multiplies the pressure drop directly, $\Delta p/\Delta p_0 = (\mu/\mu_0)/(K/K_0)$, and is a formulation input rather than a quantity hidden inside the geometric permeability $K(\varepsilon_{\mathrm{s}})$. Ranking every lever by the fractional shift of its primary model coordinate then answers which uncertainty actually moves the boundary: over the tested ranges the transport group (δ/D_{eff}, D_{eff}) produces the largest saturation-marker span, the solubility reference follows, and the production accessibility-closure parameters move $Q_{\mathrm{f,cal}}$; viscosity is the largest *pressure* lever, while the smooth $K(\varepsilon_{\mathrm{s}})$ response is negligible over the modeled load. Together these sweeps turn the scope bounds (scope-and-bounds table in the SI) into quantified sensitivities rather than asserted caveats. Full per-family values and evidence classes are reported in the Supplementary Information.

4.6 A conditional operating map and external dissolution scales

The iodine-state progression projects into an operating boundary in the current-density/areal-capacity plane (Fig. 8). Both the saturation boundary, $J_s^+(Q)$, and the production half-accessibility boundary, $J_{b,\mathrm{cal}}^+(Q)$, are conditional on the registered parameterization. Converged marker solves at $J = 20$ and 40 $\mathrm{mA\,cm^{-2}}$ anchor the local interpolation. The registered $J = 40$ trajectory also supplies $Q_{\mathrm{f,cal}}$. No physical-island failure boundary is inferred.

The map carries conditional model magnitudes and external kinetic context; it does not turn them into one validated ceiling. Published iodine-dissolution measurements provide an external kinetic scale rather than a calibrated ceiling for the present electrolyte. Zhao *et al.* reported method-dependent fluxes corresponding to approximately 122 (chronoamperometry) and 232 $\mathrm{mA\,cm^{-2}}$ (UV–visible) at 1 M iodide, and 425 (chronoamperometry) and 984 $\mathrm{mA\,cm^{-2}}$ (UV–visible) at 2 M iodide [19]. The former 232–425 $\mathrm{mA\,cm^{-2}}$ span combined different concentrations and methods and is not retained as a same-condition band. These values motivate a clearance-versus-generation hypothesis but do not independently validate a sustainable charge-current ceiling for the present ZnI_2/NH_4Br cell. Likewise, the charge-side production marker and measured maximum discharge capacities are different observables and are not equated.

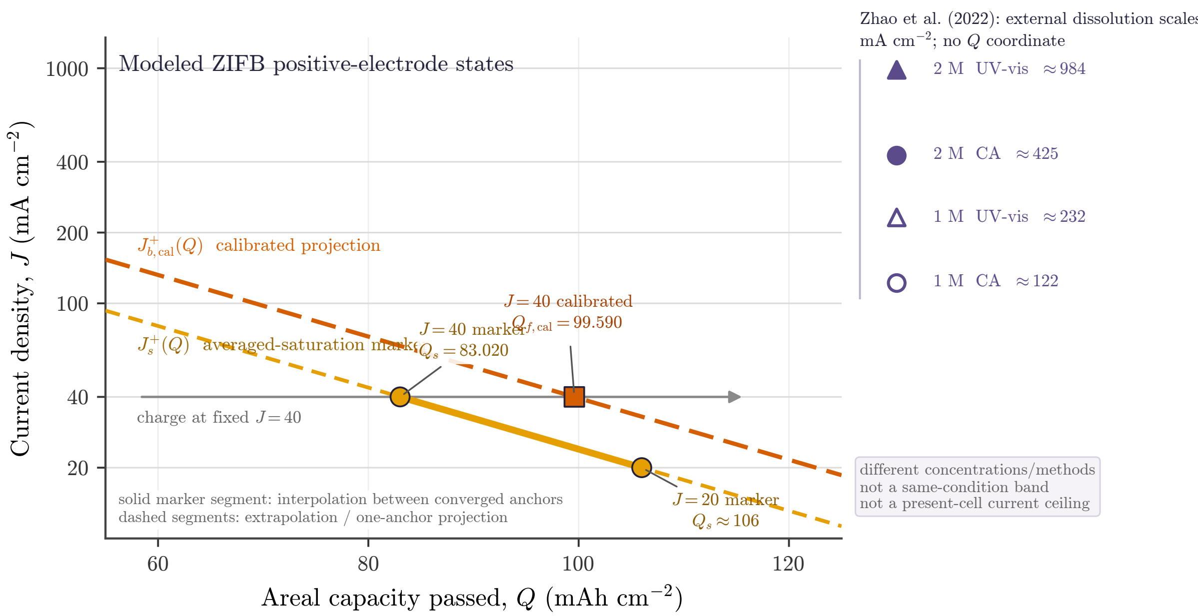

Figure 8: Organizing map for the modeled ZIFB positive-electrode states. At fixed current, increasing Q crosses the modeled saturation boundary $J_s^+(Q)$ and, on the production closure, the calibrated half-accessibility projection $J^+_{b,\mathrm{cal}}(Q)$. Circles are converged saturation-marker solves at $J = 20$ and $40\ \mathrm{mA\,cm^{-2}}$; the square is the registered $J = 40$ marker $Q_{\mathrm{f,cal}} = 99.590\,\mathrm{mAh\,cm^{-2}}$. The solid saturation-marker segment is the interpolation between the two converged anchors. Dashed saturation-marker segments are extrapolations, and the dashed $J^+_{b,\mathrm{cal}}$ curve is a parallel one-anchor projection rather than a solved boundary. The separate right-hand rail lists the four concentration- and method-specific dissolution-derived scales reported by Zhao *et al.* [19]; the rail has no Q coordinate and is neither a same-condition band nor a current ceiling for the present cell. Source data are provided with the figure package.

Literature anchors for the clearance hypothesis. The relation $i^* = 2Fk$ translates reported heterogeneous I_2-dissolution fluxes into a condition-labelled current scale; it is not fitted to the present cells. Accelerating dissolution with a cosolvent increases that scale [19]. Independent literature also supports positive-electrode accessibility and kinetic penalties in iodine-precipitation regimes: a dense I_2 film raises the positive charge-transfer resistance roughly ninefold ($16.5 \rightarrow 147.4\ \Omega\,\mathrm{cm}^2$) [18], and positive-electrode catalysts or accelerated iodine dissolution can improve high-rate behavior [39, 45]. These examples are mechanistic context, not validation of a common ceiling or of the present thresholds.

Representative supporting-electrolyte context. The positive-electrode scaffold is evaluated at a representative 1 M ZnI_2+4 M NH_4Br condition. In the rebuilt common-current full-cell series, the maximum of the within-command-level median delivered capacity increased descriptively from 40.84 $\mathrm{mAh\,cm^{-2}}$ at 0 M to 137.54 $\mathrm{mAh\,cm^{-2}}$ at 4 M NH4Br. Independent-cell counts were 1/1/1/2/1 for 0/1/2/3/4 M, respectively. This protocol-bounded association neither establishes a population fold change or saturation law nor validates the continuum thresholds. The complete composition trajectories, selection rules and endpoint sensitivities are reported in the Supplementary Information. NH4Br is retained here only as the representative supporting electrolyte; the paper's mechanistic claims concern the ZIFB positive electrode.

4.7 Bounding the alternative failure channels

We bound two alternatives rather than excluding all transport or film mechanisms. *Not a smooth geometric permeability collapse*: the relative permeability stays near unity throughout (Kozeny–Carman endpoint 0.96, pore-network 0.98; $K/K_0 = 0.998$ at $\langle\theta_{\mathrm{cal}}\rangle = \frac{1}{2}$ on the production trajectory), so injecting the physical pore-network closure in place of the mean-field one shifts the cell voltage by only 1.27 mV (permeability-injection figure in the SI). The full two-way Darcy flow is likewise immaterial to the solved saturation marker (with the δ sub-grid caveat of §4.8): the solved velocity field is nearly invariant ($<0.1\%$ over a charge), and a separate frozen-velocity control leaves the registered saturation marker unchanged. This isolates the *geometric* permeability channel and does not remove transport from the mechanism: the free-I_2 generation stress $\Pi_{\mathrm{gen}} = i_{\mathrm{app}}\delta/(2FD_{\mathrm{eff}}c_{\mathrm{sat}})$ remains central to the super-saturation marker. Discrete local pore-throat/shell blockage and viscosity-driven pressure ($\Delta p \propto \mu/K$) lie outside what the near-unity geometric $K(\varepsilon_{\mathrm{s}})$ tests (Supplementary Table S14). Within the registered production parameterization, the modeled failure coordinate is loss of *accessible reaction sites*, not a smooth geometric permeability collapse. The far-extrapolated hydraulic response as $K/K_0 \rightarrow 0$ is a distinct downstream scenario outside the validated range and present scope. *Voltage is morphology-non-identifying*: the terminal voltage does not uniquely identify deposit morphology. In the reduced observation layer, accessibility loss and the Tafel-like coefficient trade off over a broad range, so the same voltage trend can be represented by different effective accessibility states. The physical isolated-island shadow calculations bound one geometric response, while the production relation is an effective voltage-calibrated response. In the registered matched solves, replacing only the accessibility relation changes the terminal voltage by -288.156 mV and changes the coupled solid-inventory trajectory (§4.4). Their difference shows model-form sensitivity, not proof of a coalesced film. The current evidence therefore supports only the following statement: under the fixed production parameterization, a stronger effective accessibility-loss response than the tested isolated-island relation is needed to reproduce the calibrated late-rise branch, but the responsible morphology remains unresolved. The smooth bulk-permeability calculation rules out only a mean-field hydraulic collapse over the modeled load and does not decide among local pore-throat, shell, island-coalescence or other accessibility-loss mechanisms. As external context, a dense iodine film has been reported to raise positive-electrode charge-transfer resistance approximately ninefold [18]; a uniform area-blocking translation would correspond to an effective accessibility loss near 0.89, but this mapping is not a morphology measurement for the present cell.

Beyond these internal alternatives, three *external* channels can also close a ZIFB operating window, and the present mechanism is bounded against each rather than claimed to supersede them. First, the negative zinc electrode: two ZIFB degradation studies attribute their reported performance limit to a dense zinc layer or dendrite front at the anode [8, 46]. Those reports identify negative-side degradation in their cells; this is complementary to, but does not exclude, positive-side limitations under other operating conditions (§5.1). Second, altered speciation or storage pathways: quaternary-ammonium chemistry can form an insoluble iodine-containing salt, while halide mediation can redirect oxidized-iodine speciation [47, 48]; neither regime is represented by the present unmediated I_2/I^- branch. Third, adsorptive site passivation on catalytic hosts can dominate on strongly binding carbons [49]; the felt-class carbon modeled here carries no such catalyst. Each competing channel therefore sharpens, rather than contradicts, the scope of the saturation–deposition–accessibility positive-electrode scaffold.

4.8 Operating and design levers act on three nodes

Every global operating variable examined here acts on one of the terms in $S_{\mathrm{base}} = \gamma_{\mathrm{salt}}(\Pi_{\mathrm{inv}} + \Pi_{\mathrm{gen}})/\beta_{\mathrm{intra}}$ (§2): the free-I_2 generation stress $\Pi_{\mathrm{gen}} = i_{\mathrm{app}}\delta/(2FD_{\mathrm{eff}}c_{\mathrm{sat}})$ is moved by rate, the oxidized-carrier diffusivity, mechanical compression and the intrinsic reference c_{sat} (through $c_{\mathrm{sat}}^{\mathrm{eff}}$ at fixed γ_{salt}); the retained-inventory stress $\Pi_{\mathrm{inv}} = c_{\mathrm{I_2,tot}}/c_{\mathrm{sat}}$ by the state of charge; and the within-cycle buffer β_{intra} by the supporting electrolyte. Conductivity, by contrast, moves the cell voltage but leaves the saturation marker pinned, a clean negative control. The solved flow rate also leaves that marker pinned, but partly *by construction*: the fiber-scale boundary layer δ that carries the mass-transfer stress is a sub-grid parameter decoupled from the electrode-scale Darcy field, so the solve does not capture the physical route by which a higher flow rate thins δ and lowers Π_{gen}. That route is instead quantified analytically by the δ/D_{eff} sweep in the Supplementary Information. Ranking every lever by the node it perturbs and its effect on S places the oxidized-carrier diffusivity as the dominant lever through $\Pi_{\mathrm{gen}} \propto 1/D_{\mathrm{eff}}$ (Table 3; the per-family COMSOL scans and a phase-plane view, in which each lever is a distinct move on one saturation boundary, are in the Supplementary Information).

Three tiers emerge (Table 3). The largest raw effects, the oxidized-carrier diffusivity prior and the salting-out solubility, set the absolute failure scale but rest on the least-constrained inputs, so their *ranking*, not any calibrated magnitude, is load-bearing. The leading operationally accessible, clean lever is the applied current density. Conductivity forms an orthogonal voltage channel that leaves the saturation marker pinned (the clean negative control). The solved flow also leaves that marker pinned; because the fibre boundary layer δ is sub-grid, we test this analytically with a flow-dependent $\delta(v)$ scenario from a Sherwood correlation ($Sh = a\,Re^m Sc^{1/3}$, $\delta \propto v^{-m}$; Fig. 9): across the operating range 25–100 $\mathrm{mL\,min^{-1}}$ the marker moves only $\approx 11\,\mathrm{mAh\,cm^{-2}}$ (Sherwood exponent $m = 0.4$–0.6), against $\approx 61\,\mathrm{mAh\,cm^{-2}}$ for the D_{eff} and rate levers, so flow remains the weaker saturation-marker lever within this declared boundary-layer scenario. These model trends are juxtaposed with descriptive, single-cell compression capacity brackets in the SI; those measurements do not independently validate the modeled trend.

Design implication. The registered production branch places its saturation and calibrated half-accessibility markers at 83.0 and 99.6 $\mathrm{mAh\,cm^{-2}}$, respectively. These conditional model coordinates are used to compare lever directions, not as demonstrated electrode limits. Operationally,

Table 3: Model-internal operating and design levers for the ZIFB positive electrode. Each row states the model channel, directional response and evidence class. Numerical sweep details are given in the Supplementary Information.

Knob / family	Model channel	Directional response	Evidence class
Oxidized-carrier diffusivity	$\Pi_{\text{gen}} \propto 1/D_{\text{eff}}$	Lower D_{eff} advances Q_{s} and increases the production solid/accessibility response.	bounded MD prior; solved sweep
Apparent iodine solubility	$\Pi_{\text{inv}}, \Pi_{\text{gen}} \propto 1/c_{\text{sat}}$	Lower c_{sat} reduces headroom on both stress terms and advances Q_{s}.	bounded literature prior
Applied current density	$\Pi_{\text{gen}} \propto i_{\text{app}}$	Higher current raises the generation stress and advances modeled saturation.	solved points plus interpolation
Mechanical compression	εL and D_{eff}	Compression changes pore inventory and transport; the available cells test direction only.	solved model; descriptive data
Charge passed / iodine inventory	$\Pi_{\text{inv}}(Q)$	Charging increases inventory stress along the registered trajectory.	solved state coordinate
Representative supporting electrolyte	β_{intra} and unsolved transport/speciation inputs	NH4Br sets the baseline condition; cross-composition effects are not separated.	prescribed baseline; descriptive data
Wetted area a_s	local current distribution, not global Π_{gen}	One branch changed current crowding; the sign is not identified.	one branch only
Conductivity and flow	voltage / boundary-layer channels	Conductivity changes voltage without moving S; flow acts through unresolved $\delta(v)$ in the reduction.	control and analytic bound

halving the applied current from 40 to 20 mA cm^{-2} moves the saturation marker from ≈ 83 to $\approx 106\,\text{mAh cm}^{-2}$, a $\sim$28% increase in the modeled marker capacity. On the formulation side the oxidized-carrier diffusivity is the dominant and most *asymmetric* lever. In the solved sweep (§4.5), doubling it delays the marker to $\approx 106\,\text{mAh cm}^{-2}$ (+28%—the same gain as halving the current, since both act only through $\Pi_{\text{gen}} \propto i_{\text{app}}/D_{\text{eff}}$), whereas a slower carrier is far more costly: halving the diffusivity drops the marker to $\approx 45\,\text{mAh cm}^{-2}$ (-46%). Fast oxidized-iodine transport is therefore something to protect, not to trade away for other electrolyte properties. Raising the apparent solubility buys headroom on both stress channels, while faster dissolution increases the external clearance scale $2Fk$ [19]. The model therefore prioritizes lowering the free-I_2 generation stress and improving clearance over increasing inventory alone.

5 Discussion

5.1 Two electrodes, two regimes

The solved contribution is confined to the ZIFB positive electrode. Negative-electrode transport and morphology remain external scope boundaries; the present evidence does not establish a universal division in which one electrode always limits capacity and the other always limits rate. On the positive side, the modeled native-solid deposition regime is distinct from dissolved-iodine shuttle [12, 13] and high-bromide interhalogen chemistry [48, 50, 51]. Within the solved regime we keep four objects separate: the saturation marker $S = 1$, retained inventory ε_{s}, a branch-specific accessibility state (θ_{cal} or θ_{island}), and smooth hydraulic permeability K. The interval ΔQ_{cal} is a production-model coordinate, not a directly measured reservoir or a threshold transferred to the negative

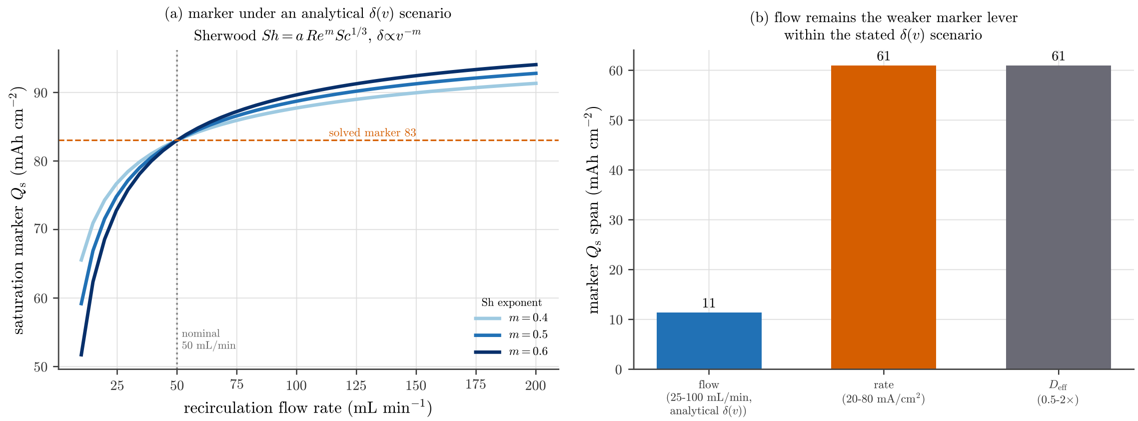

Figure 9: Flow remains a weak saturation-marker lever in an analytical boundary-layer scenario. The solved Darcy flow leaves the marker pinned partly because the fibre Nernst layer δ is a sub-grid parameter; here δ is made flow-dependent through a Sherwood correlation ($Sh = a\,Re^m Sc^{1/3}$, so $\delta \propto v^{-m}$), anchored to $\delta = 25\,\mu\text{m}$ at the nominal $50\,\text{mL}\,\text{min}^{-1}$, and only the generation stress Π_{gen} scales with flow while $\Pi_{\text{inv}}(Q)$ and $\beta(Q)$ are read from the solved baseline. (a) Saturation marker Q_s versus recirculation rate for Sherwood exponents $m = 0.4$–0.6. (b) Over the operating range 25–100 $\text{mL}\,\text{min}^{-1}$ the marker shifts by only $\approx 11\,\text{mAh}\,\text{cm}^{-2}$, against $\approx 61\,\text{mAh}\,\text{cm}^{-2}$ for the D_{eff} and rate levers. This declared correlation bounds, but does not measure, the unresolved sub-grid $\delta(v)$ channel.

electrode.

5.2 The negative branch as a scope boundary

An illustrative diffusion-limit estimate gives CDR $= j/j_{\text{lim}} = 0.07$–$0.2$ over 40–100 $\text{mA}\,\text{cm}^{-2}$ under the declared $c_{\text{Zn}} = 1\,\text{M}$ and $\delta = 25\,\mu\text{m}$ assumptions (SI). Because those inputs are scenario values rather than measured negative-electrode boundary conditions, the estimate does not establish the actual transport margin. Negative-side morphology and separator failure (the documented compact-layer/perforation channel, whose controlling current can run *opposite* in sign to the positive supersaturation [8, 10]) are therefore treated as external capacity-boundary risks rather than solved rate limits. This estimate does not resolve dendrite morphology or interfacial kinetics, so the negative branch is carried only as a literature-anchored regime analogue.

5.3 Supporting-electrolyte scope

The positive-electrode mechanism is formulated for a fixed representative electrolyte. Supporting-electrolyte composition can change speciation, apparent iodine accommodation, bulk transport, conductivity and viscosity, but the present full-cell composition series does not identify those contributions separately. We therefore treat NH4Br as the representative supporting electrolyte used for baseline parameterization and report its composition trend descriptively in the Supplementary Information. Bromide is not oxidized in the modeled I_2/I^- branch and is not the subject of the paper. Literature supports roles for bromide complexation and distinguishes some $NH4^+$ and Br^- effects [33–35], but those mechanisms are not independently resolved by the present cells. Extension to another electrolyte requires independent speciation and transport inputs; the present composition data do not define a transferable bromide coefficient.

5.4 Testable predictions

The registered model makes four falsifiable, electrode-level statements; each retains its model conditions:

1. **The averaged-saturation and calibrated-accessibility markers are separated on the production branch.** At $J = 40\,\mathrm{mA\,cm^{-2}}$, the model gives $Q_\mathrm{s} = 83.0$ and $Q_\mathrm{f,cal} = 99.6\ \mathrm{mAh\,cm^{-2}}$. Simultaneous, electrode-resolved measurement of free iodine and an independent accessibility observable can test this ordering; a full-cell capacity endpoint cannot.

2. **The saturation marker advances with the generation Damköhler number.** The modeled Q_s moves to lower capacity as $\Pi_\mathrm{gen} \propto \delta/D_\mathrm{eff}$ rises—at higher current, thicker boundary layer, or lower oxidized-carrier diffusivity in the registered sweeps—advancing roughly linearly over the tested range. Published $2Fk$ values supply condition-specific clearance scales rather than a universal threshold for this test.

3. **The calibrated model predicts a downstream voltage-derivative signature.** The production trajectory places its modeled $\mathrm{d}V/\mathrm{d}Q$ maximum after $Q_\mathrm{f,cal}$. The existing full-cell derivatives are sensitive to cycle and smoothing window and do not validate this ordering: in the all-eligible-cycle reconstruction, most detected rise onsets precede Q_s and the maxima fall on both sides of $Q_\mathrm{f,cal}$. The statement is therefore retained only as a prospective electrode-resolved test.

4. **Voltage alone cannot select the deposit morphology.** Independent solid-inventory and spatial accessibility measurements must distinguish the production relation from the physical isolated-island family. If the same measured inventory and morphology-resolved accessibility follow the dense-island coupled trajectory while the voltage follows the production trajectory, the present closure set is incomplete.

Candidate protocols include calibrated in-situ UV–visible quantification of dissolved iodine species [52, 53] and a retained-iodine mass balance proposed here. Optical relaxation after electrodeposition has precedent in a distinct $ZnCl_2$ water-in-salt/FTO electrochromic device [28]. Dissolved-species concentration imaging has precedent in a nonaqueous TEMPO/$\mathrm{TEMPO^+}$ flow battery [54], while impedance can track iodine-film interfacial resistance [18]. These are prospective method analogues, not validation of the present felt or its closure. No additional physical experiment is used in the present study.

5.5 Scope and bounds

The continuum calculation is a single-charge positive-electrode mechanism scaffold, not an exact voltage validator or a cycle-life model. Its registered production accessibility relation is calibrated, whereas the isolated-island and pore-network calculations are physical bounds; no deposit morphology is identified. The chemistry is the unmediated I_2/I^- branch in a felt-class porous carbon under the stated ZnI_2/NH_4Br condition. Nano-confined solid-iodine hosts [16], dry ultra-high-loading electrodes [7], insoluble quaternary-ammonium iodine salts and interhalogen pathways [47, 48] follow different storage or reaction rules. Deep hydraulic runaway, negative-electrode morphology, discharge redissolution and cycle-to-cycle reset are also outside the solve. Published dissolution kinetics [19] provide external scale context, not validation of the present current or capacity markers. The existing full-cell datasets are reported descriptively and do not close the morphology or electrode-resolved validation gap.

6 Conclusion

We developed an auditable state model for the ZIFB positive electrode that separates free-iodine saturation, retained solid inventory and modeled accessibility loss. Within the registered production branch, $Q_s = 83.0\,\mathrm{mAh\,cm^{-2}}$ precedes the voltage-calibrated half-accessibility marker $Q_{\mathrm{f,cal}} = 99.6\,\mathrm{mAh\,cm^{-2}}$ at $40\,\mathrm{mA\,cm^{-2}}$. Their absolute positions and separation are closure- and prior-dependent model outputs. Fresh matched continuum solves show that changing only the accessibility relation alters both the solid-inventory trajectory and terminal voltage by 288.2 mV on the convergence-qualified true mesh, demonstrating model-form sensitivity without identifying the real deposit morphology. The physical isolated-island calculations and smooth pore-network permeability calculation bound possible mechanisms but do not validate the production closure. Published dissolution kinetics provide external scale for a clearance-versus-generation hypothesis, not a current ceiling for the present electrolyte. The principal contribution is therefore a traceable positive-electrode accounting with explicit falsification tests for each link. NH4Br is only the representative supporting electrolyte used in the present parameterization.

Methods

Density-functional theory. Periodic DFT used CP2K/Quickstep [55, 56], the PBE functional with D3(BJ) dispersion [57–59], and a DZVP-MOLOPT-SR-GTH basis [60] with GTH pseudopotentials [61–63]; the plane-wave density cutoff was 150 Ry, the relative cutoff 30 Ry, and sampling used the Γ point on graphitic slabs and ribbons with oxygenated defects. Geometries were optimized, single-point energies were counterpoise-corrected for basis-set superposition error [64], and open-shell sites (single vacancy) treated spin-polarized (LSD, Fermi smearing, 80 added states). Single-I_2 adsorption energies referenced to gas-phase I_2 (−0.40 basal, −0.85 carbonyl, −1.00 hydroxyl, −0.32 vacancy eV) and an optional $2I_2$ coalescence (−0.69 eV at C−OH) are used as bounded mechanism priors; finite-cluster water-pair energies are retained as diagnostics only and excluded from parameterization and thermodynamic inference. **Molecular dynamics.** Classical force-field MD used GROMACS [65] for $1\,\mathrm{M}\,ZnI_2 + 4\,\mathrm{M}\,NH_4Br$ in SPC/E water [66], with explicit oxidized carriers (neutral two-site I_2; linear I_3^-/I_2Br^-). Monovalent-halide parameters followed Joung–Cheatham [67]; the Zn^{2+}, tetrahedral NH_4^+ and oxidized-iodine models were project-specific SPC/E-compatible or analogue parameterizations whose numerical values are frozen in the archived topologies and are not treated as validated force fields. Packmol-built boxes [68], were energy-minimized, equilibrated for 0.1 ns in NVT and 0.5 ns in NPT, and propagated for one 20 ns NVT production trajectory per SOC/charge case at 298.15 K with GROMACS 2026.0-conda_forge. Diffusivities were obtained from center-of-mass mean-squared displacements over a 2.51 ns to 8.49 ns lag window. The displayed block statistic is the standard deviation of four contiguous 5 ns time-block estimates divided by two; it diagnoses within-trajectory stability and is not an independent-replica standard error. GFN2-xTB/ALPB-water calculations supplied only polyhalide charge priors [69]; bonded and Lennard–Jones terms remain project analogues. Electronic-continuum charge scaling ($q \approx 0.8$) [70] bracketed by formal charges ($q = 1.0$). All 18 planned viscosity-replica runs lack energy outputs and are excluded; the MD result is reported only as a force-field-limited transport prior. **Mesoscale closures.** Cylindrical fiber unit cell ($r_f = 7.5\,\mu\mathrm{m}$, $\varepsilon = 0.85$) with cylindrical finite-difference diffusion ($8 \times 36 \times 36$) for the physical $\theta_{\mathrm{island}}(\varepsilon_s)$ family; 15^3 pore-throat network (lognormal radii, sparse SPD solve) for $K(\varepsilon_s)$. **Continuum model.** Finite-element tertiary current distribution with two-way Darcy/porous-media flow (velocity solved and coupled into transport) in COMSOL Multiphysics 6.3 (implicit BDF time stepping). The modeled cell is a

$4\,\mathrm{cm}^2$ flow frame with a 2 mm pristine graphite felt ($\varepsilon = 0.85$), a Nafion 212 membrane, a 25 mL reservoir and $50\,\mathrm{mL\,min^{-1}}$ recirculation at 293.15 K, charged galvanostatically at $i_{\mathrm{app}} = 400\,\mathrm{A\,m^{-2}}$ in 1 M ZnI_2+4 M NH_4Br. The bare-carbon soluble-iodine path uses Butler–Volmer kinetics with $E^{0\prime} = 0.548\,\mathrm{V}$, $c_{\mathrm{ref}} = 120\,\mathrm{mol\,m^{-3}}$, $i_0 = 5 \times 10^4\ \mathrm{A\,m^{-2}}$ and $\alpha_a = \alpha_c = 1$. The native deposited-species path uses the anodic Tafel law $j_{\mathrm{dep}} = i_{0,\mathrm{dep}} 10^{\eta/A_a}$ with $i_{0,\mathrm{dep}} = 0.05\,\mathrm{A\,m^{-2}}\max(S_{\mathrm{local}} - 1, 0)$ and $A_a = 118\,\mathrm{mV}$. The free-monomer stress uses the salting-out effective solubility $c_{\mathrm{sat}}^{\mathrm{eff}} = c_{\mathrm{sat}}/\gamma_{\mathrm{salt}}$ with $\gamma_{\mathrm{salt}} = 3$ (a bounded cross-salt activity prior, not a measurement of the present electrolyte; §2.3). The exact $E^{0\prime}$, c_{ref}, exchange-current and Tafel parameters, $20\,\mathrm{S\,m^{-1}}$ liquid conductivity, 25 µm mass-transfer length and auxiliary phase-transfer rates are prescribed or empirical model values rather than independently identified properties; their roles and sensitivity are classified in the SI. A dynamic nucleation-state ODE and algebraic accessibility feedback complete the solve. The production relation $\theta_{\mathrm{cal}} = 1 - \exp[-6 N_{\mathrm{I}_2}^{1/3} \varepsilon_{\mathrm{s,reg}}^{2/3}]$ is a voltage-calibrated effective closure. The physical dense isolated-island relation of Eq. (17) is analyzed separately. Fresh matched control and dense-island coupled calculations were generated from byte-identical copies of the canonical input (SHA-256 `4813B306F39330CCEAF819C4AB8815C55A3DF8A04726741F33ACB801B8806B9B`), with study `stdR522`, live solution `sol5`, separate output datasets and the same time grid; only the two registered accessibility expressions were changed. The pore-network closure tests smooth bulk permeability only. DFT and MD outputs are bounded priors rather than fitted kinetic constants. Central inputs are tabulated in the SI; the complete machine-readable parameter inventory, model identities and output hashes are included in the release repository. **Reduced model.** Eq. (10) was fit by linear least squares in its observation-layer parameters to an existing pre-failure charge trace where explicitly reported. **Existing experimental data.** The reconstructed composition, rate and compression reconstructions are descriptive full-cell analyses at the physical-cell level; the G4 derivative analysis is a one-cell smoothing-and-cycle sensitivity analysis that does not validate model ordering. None is described as independent validation of the production thresholds.

Data and code availability

Processed source tables, deterministic analysis scripts, model-branch manifests and per-figure provenance records will be deposited in the repository identified in the final submission metadata. The release record will identify the canonical COMSOL input hash, study, solution, dataset, software version and exported traces; no original solved model is overwritten. Legacy acquisition filenames and condition fields containing "NH4Cl" are operator-confirmed metadata-entry errors: all affected experiments used NH4Br. Raw files, filenames and SHA-256 hashes are preserved unchanged, while the analysis layer maps the legacy label to NH4Br under correction ID EXP-META-001. The path-level and unique-hash correction manifests are included with the processed experimental data. Repository URLs and DOIs will be stated in the present tense only after the records exist.

Author contributions

The final CRediT statement must be inserted from the verified author list before submission; roles and approval cannot be inferred from the scientific files.

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Competing interests

The final competing-interest statement must be confirmed by every author before submission.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During manuscript preparation, the authors used OpenAI Codex/ChatGPT for language revision, consistency checking, code-assisted data auditing, code review and assistance drafting deterministic analysis and plotting scripts. The authors reviewed and edited all generated material, verified calculations and citations against the underlying sources, and take full responsibility for the manuscript. Quantitative artwork was rendered from archived numerical data by documented scripts; no generative image synthesis or AI-based alteration of primary research images was used. No AI system is listed as an author.

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